

Stochastic Analysis on Manifolds

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Chapter 1

SDE and Diffusion

Fix probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ of σ -fields contained in $\mathcal{F}$ with assumption that $\mathcal{F}_\infty = \mathcal{F}$. Also, $\mathbb{F}$ satisfies the usual condition, i.e., complete and right-continuous. Without specification, all semimartingales are assumed continuous. We use Einstein summation convention for convenience.

1.1 SDE on Euclidean Space

Solution of SDE: Let $Z = (Z_t)_{t \geq 0}$ be a continuous $\mathbb{R}^\ell$ -valued $\mathbb{F}$ -semimartingale, i.e.,

$$Z = M + A,$$

where M is a local martingale and A is an adapted process of local bounded variation such that $A_0 = 0$. Let $\sigma: \mathbb{R}^N \rightarrow \mathbb{R}^{N \times \ell}$ be smooth and locally Lipschitz, i.e. for all $R > 0$ there exists $R > 0$ and $C(R) > 0$ such that

$$\|\sigma(x) - \sigma(y)\|_{\mathbb{F}} \leq C(R) \|x - y\|, \quad \forall x, y \in B(R).$$

Let $X_0 \in \mathbb{R}^N$ be $\mathcal{F}_0$ measurable. For a stopping time τ , consider a semimartingale $X = (X_{t \wedge \tau})_{t \geq 0}$ of the form

$$X_t = X_0 + \int_0^t \sigma(X_s) dZ_s, \quad 0 \leq t \leq \tau.$$

We say X is a solution of a SDE driving by the semimartingale Z , and such equation denoted as $\text{SDE}(\sigma, Z, X_0)$. Moreover, by Itô formula, for $f \in C^2(\mathbb{R}^N)$,

$$\begin{aligned} f(X_t) &= f(X_0) + \int_0^t f_{x_i}(X_s) \sigma_\alpha^i(X_s) dZ_s^\alpha \\ &\quad + \frac{1}{2} \int_0^t f_{x_i x_j}(X_s) \sigma_\alpha^i(X_s) \sigma_\beta^j(X_s) d\langle Z^\alpha, Z^\beta \rangle_s \\ &= f(X_0) + \int_0^t f_{x_i}(X_s) \sigma_\alpha^i(X_s) dM_s^\alpha + \int_0^t f_{x_i}(X_s) \sigma_\alpha^i(X_s) dA_s^\alpha \\ &\quad + \frac{1}{2} \int_0^t f_{x_i x_j}(X_s) \sigma_\alpha^i(X_s) \sigma_\beta^j(X_s) d\langle M^\alpha, M^\beta \rangle_s \end{aligned}$$

Remark 1.1.1. Note that if $X_0 \in L^2$ and σ is globally Lipschitz continuous, then we don't need the stopping time τ and $\text{SDE}(\sigma, Z, X_0)$ has a unique solution $X = (X_t)_{t \geq 0}$.

The problem why we need to consider a stopping time is because of the local Lipschitz property of σ . The local Lipschitz would make the usual solution exploded. To consider it rigorously, we first need the definition of explosion time.

Let M be a locally compact metric space and $\widehat{M} = M \cup \{\partial_M\}$ be its one-point compactification. A continuous map $x: [0, \infty) \rightarrow \widehat{M}$ has an explosion time $e = e(x) > 0$ if $x_t \in M$ for $0 \leq t < e$ and $x_t = \partial_M$ for $t \geq e$ when $e < \infty$, i.e.

$$e: W(M) \rightarrow [0, \infty]$$

where $W(M) \subset C([0, \infty), \widehat{M})$ such that e can be defined. Moreover, $W(M)$ can be equipped with the canonical filtration $\mathbb{F}_M = (\mathcal{F}_{M,t})_{t \geq 0}$, i.e., the natural filtration generated by coordinate processes. Then e is a $\mathbb{F}_M$ -stopping time.

Definition 1.1.2. Let τ be a stopping time. A continuous $X = (X_{t \wedge \tau})_{t \geq 0}$ is called a semimartingale up to τ if there exists a sequence of stopping times $\tau_n \uparrow \tau$ such that $X^{\tau_n} = (X_{t \wedge \tau_n})$ is a semimartingale.

Definition 1.1.3. A semimartingale X up to a stopping time τ is a solution of $\text{SDE}(\sigma, Z, X_0)$ if there exists a sequence of stopping times $\tau_n \uparrow \tau$ such that $X^{\tau_n} = (X_{t \wedge \tau_n})$ is a semimartingale and

$$X_{t \wedge \tau_n} = X_0 + \int_0^{t \wedge \tau_n} \sigma(X_s) dZ_s, \quad t \geq 0.$$

Theorem 1.1.4. If σ is locally Lipschitz continuous, then there exists a unique $W(\mathbb{R}^N)$ -valued random variable X which is a solution of $\text{SDE}(\sigma, Z, X_0)$ up to its explosion time $e(X)$. Moreover, if X, Y are two solutions up to τ and η respectively, then $X_{t \wedge \tau \wedge \eta} = Y_{t \wedge \tau \wedge \eta}$. In particular, if X is a solution up to $e(X)$, then $\eta \leq e(X)$ and $X_{t \wedge \tau} = Y_{t \wedge \tau}$.

Theorem 1.1.5. Suppose σ and b are locally Lipschitz. Then the weak uniqueness holds for

$$X_t = X_0 + \int_0^t \sigma(X_s) dW_s + \int_0^t b(X_s) ds,$$

where W is a Brownian motion.

Stratonovich integral. For two continuous semimartingales H, Z , the Stratonovich integral is defined as

$$\int_0^t H_s \circ dZ_s = \int_0^t H_s dZ_s + \frac{1}{2} \langle H, Z \rangle_t.$$

In such case, the Itô formula becomes

$$df(X_t) = \frac{\partial f}{\partial x^\alpha}(X_t) \circ dX_t^\alpha.$$

Suppose that $V_\alpha, \alpha = 1, 2, \dots, \ell$ are smooth vector fields in $\mathbb{R}^N$, that is,

$$V_\alpha = (V_\alpha^1, \dots, V_\alpha^N): \mathbb{R}^N \rightarrow \mathbb{R}^N$$

and vector fields means that for any $f \in C^1(\mathbb{R}^N)$,

$$V_\alpha f = \sum_{i=1}^N V_\alpha^i \frac{\partial f}{\partial x^i}.$$

More generally, if

$$X_t = X_0 + \int_0^t V_\alpha(X_s) \circ dZ_s^\alpha,$$

then

$$X_t = X_0 + \int_0^t V_\alpha(X_s) dZ_s^\alpha + \frac{1}{2} \int_0^t \nabla_{V_\beta} V_\alpha(X_s) d\langle Z^\alpha, Z^\beta \rangle_s.$$

Proof. By definition,

$$dX_t = V_\alpha(X_t) \circ dZ_t^\alpha = V_\alpha(X_t) dZ_t^\alpha + \frac{1}{2} d\langle V_\alpha(X_t), Z_t^\alpha \rangle.$$

or

$$dX_t^i = \sum_{\alpha=1}^{\ell} V_\alpha^i(X_t) dZ_t^\alpha + \frac{1}{2} \sum_{\alpha=1}^{\ell} d\langle V_\alpha^i(X_t), Z_t^\alpha \rangle.$$

For $d\langle V_\alpha^i(X_t), Z_t^\alpha \rangle$, first because $\sum_{\alpha} V_\alpha^i(X_t) dZ_t^\alpha$ is the local martingale part of dX_t^i ,

$$d\langle X^i, Z^\alpha \rangle_t = \sum_{\beta=1}^{\ell} V_\beta^i(X_t) d\langle Z^\beta, Z^\alpha \rangle_t.$$

Then by Itô formula,

$$dV_\alpha^i(X_t) = \sum_{j=1}^n \frac{\partial V_\alpha^i}{\partial x^j} dX_t^j + \text{bounded variation part}.$$

So

$$\begin{aligned} d\langle V_\alpha^i(X_t), Z_t^\alpha \rangle &= \sum_{j=1}^n \frac{\partial V_\alpha^i}{\partial x^j} d\langle X^j, Z^\alpha \rangle_t \\ &= \sum_{\beta=1}^{\ell} \sum_{j=1}^n V_\beta^j \frac{\partial V_\alpha^i}{\partial x^j} d\langle Z^\beta, Z^\alpha \rangle_t \\ &= \nabla_{V_\beta} V_\alpha^i d\langle Z^\beta, Z^\alpha \rangle_t. \end{aligned}$$

Therefore,

$$dX_t = V_\alpha(X_t) dZ_t^\alpha + \frac{1}{2} \nabla_{V_\beta} V_\alpha(X_t) d\langle Z^\beta, Z^\alpha \rangle_t. \quad \square$$

Remark 1.1.6. Let X, Y, Z, Y, H, K are be $\mathbb{R}$ -semimartingales.

(1) Note that, if $dX = H \circ dW$ and $dY = K \circ dZ$, because

$$H \circ dW = HdW + \frac{1}{2} \langle H, W \rangle, \quad K \circ dZ = KdZ + \frac{1}{2} \langle K, Z \rangle$$

and $\langle H, W \rangle$ and $\langle K, Z \rangle$ are of bounded variation, then

$$d\langle X, Y \rangle = HKd\langle W, Z \rangle.$$

(2) In fact, we have

$$H \circ d\left(\int X \circ dW\right) = (HX) \circ dW,$$

because

$$d\langle HX, W \rangle = Hd\langle X, W \rangle + Xd\langle H, W \rangle,$$

which is because of Itô formula, $d(HX) = HdX + XdH + d\langle H, X \rangle$.

In the sense of Stratonovich integral, if $X_t = X_0 + \int_0^t V_\alpha(X_s) \circ dZ_s^\alpha$ and $f \in C^2(\mathbb{R}^N)$, then

$$f(X_t) = f(X_0) + \int_0^t (V_\alpha f)(X_s) \circ dZ_s^\alpha.$$

For a Itô SDE,

$$dX_t = b(X_t)dt + \sigma_n(X_t)dW_t^n,$$

where $W = (W^1, \dots, W^\ell)$ is a ℓ -dimensional Brownian motion. Let

$$Z_t^1 = t, \quad Z^2 = W^1, \dots, Z_{\ell+1} = W^\ell.$$

Clearly, they are semimartingales. Then by above, we can see if let

$$V_1(x) = b(x) - \frac{1}{2}\sigma_n^i \frac{\partial \sigma_n}{\partial x^i}, \quad V_2(x) = \sigma_1(x), \dots, V_{\ell+1}(x) = \sigma_\ell(x),$$

the Itô SDE is equivalent to

$$dX_t = V_\alpha(X_t) \circ dZ_t^\alpha = V_1(X_t)dt + V_\beta(X_t) \circ dZ_t^\beta.$$

1.2 SDE on Manifolds

Definition 1.2.1. Let M be a smooth manifold. A continuous M -valued process X is called a M -valued semi-martingale if $f(X)$ is a real-valued semi-martingale for all $f \in C^\infty(M)$.

Remark 1.2.2. We can also define it up to a stopping time τ as before. By Itô formula, it is obviously that $M = \mathbb{R}^N$ gives the usual definition of semimartingales. Moreover, the test function space can be chosen as $C_c^\infty(M)$.

Let $V_1, \dots, V_\ell \in \Gamma(TM)$, Z be an $\mathbb{R}^\ell$ -valued semimartingale, and an M -valued random variable $X_0 \in \mathcal{F}_0$. Consider an equation symbolically written as

$$dX_t = V_\alpha \circ dZ_t^\alpha,$$

and denoted it as $\text{SDE}(V_1, \dots, V_\ell; Z, X_0)$.

Definition 1.2.3. An M -valued semimartingale X defined up to a stopping time τ is a solution of $\text{SDE}(V_1, \dots, V_\ell; Z, X_0)$ up to τ if for all $f \in C^\infty(M)$,

$$f(X_t) = f(X_0) + \int_0^t (V_\alpha f)(X_s) \circ dZ_s^\alpha.$$

Proposition 1.2.4. Suppose $\Phi: M \rightarrow N$ is a diffeomorphism and X a solution of $\text{SDE}(V_1, \dots, V_\ell; Z, X_0)$. Then $\Phi(X)$ is a solution of $\text{SDE}(\Phi_* V_1, \dots, \Phi_* V_\ell; Z, \Phi(X_0))$ on N .

Proof. Let $Y = \Phi(X)$. For any $f \in C^\infty(N)$, because $f \circ \Phi \in C^\infty(M)$ and X is a solution of $\text{SDE}(V_1, \dots, V_\ell; Z, X_0)$,

$$f \circ \Phi(X_t) = f \circ \Phi(X_0) + \int_0^t V_\alpha(f \circ \Phi)(X_s) \circ dZ_s^\alpha.$$

Note that $\Phi_*: \Gamma(TM) \rightarrow \Gamma(TN)$ such that for and $f \in C^\infty(N)$,

$$\Phi_* V(f)(\Phi(x)) = V(f \circ \Phi)(x).$$

It follows that

$$f(Y_t) = f(Y_0) + \int_0^t \Phi_* V_\alpha(f)(Y_s) \circ dZ_s^\alpha,$$

which means that Y is a solution of $\text{SDE}(\Phi_* V_1, \dots, \Phi_* V_\ell; Z, \Phi(X_0))$. □

Note that by Whitney's embedding theorem, any smooth manifold M can be properly embedded in a Euclidean space, i.e., $i: M \hookrightarrow \mathbb{R}^N$, where i is a smooth proper map. Therefore, $i(M)$ is a closed submanifold in $\mathbb{R}^N$. Note that here closed submanifold means that a submanifold that is a closed subset. Therefore, we can always view $M \subset \mathbb{R}^N$ as an embedded submanifold that is closed. Moreover, we usually assume M without boundary. When M is non-compact, we can consider its one-point compactification $\widehat{M} = M \cup \{\partial_M\}$. So $\{x_n\}_n \subset M$ such that $x_n \rightarrow \partial_M$ in $\widehat{M}$ if and only if $\|x_n\|_{\mathbb{R}^N} \rightarrow \infty$.

In the following we always assumed that $M \subset \mathbb{R}^N$ is a submanifold that is closed and without boundary. Moreover, let

$$i = (f^1, \dots, f^N): M \hookrightarrow \mathbb{R}^N$$

and f^i are coordinate functions that $f^i(x) = x^i$.

Proposition 1.2.5. *Suppose $M \subset \mathbb{R}^N$ is a submanifold that is closed. Let $f^1, \dots, f^N$ be coordinate functions. Let X be an M -valued continuous process.*

- (1) *X is a semimartingale on M if and only if it is an $\mathbb{R}^N$ -valued semimartingale, i.e. $f^i(X)$ is a $\mathbb{R}$ -valued semimartingale for each $i = 1, 2, \dots, N$.*
- (2) *X is a solution of $\text{SDE}(V_1, \dots, V_\ell; Z, X_0)$ up to a stopping time σ if and only if for each $i = 1, 2, \dots, N$,*

$$f^i(X_t) = f^i(X_0) + \int_0^t (V_\alpha f^i)(X_s) \circ dZ_s^\alpha, \quad 0 \leq t < \sigma.$$

Proof. (1) $\Rightarrow$: It is by the definition.

$\Leftarrow$: For any $f \in C^\infty(M)$, since M is closed $\mathbb{R}^N$, it can be extended on $\tilde{f} \in C^\infty(\mathbb{R}^N)$ such that $f(X) = \tilde{f}(X)$. Because X is $\mathbb{R}$ -valued semimartingale, $f(X) = \tilde{f}(X)$ is also a semimartingale.

(2) $\Rightarrow$: It is by the definition.

$\Leftarrow$: For any $f \in C^\infty(M)$, let $\tilde{f} \in C^\infty(\mathbb{R}^N)$ be an extension. Then

$$f(X_t) = \tilde{f}(f^1(X_t), \dots, f^N(X_t)).$$

Therefore,

$$\begin{aligned} df(X_t) &= \tilde{f}(f^1(X_t), \dots, f^N(X_t)) \circ d(f^i(X_t)) \\ &= \tilde{f}(f^1(X_t), \dots, f^N(X_t)) \circ V_\alpha f^i(X_t) \circ dZ_t^\alpha \\ &= \left(\tilde{f}(f^1(X_t), \dots, f^N(X_t)) V_\alpha f^i(X_t) \right) \circ dZ_t^\alpha \\ &= V_\alpha \tilde{f}(X_t) \circ dZ_t^\alpha = V_\alpha f(X_t) \circ dZ_t^\alpha, \end{aligned}$$

where the final equality is because $V_\alpha \in \Gamma TM$ and $\tilde{f}$ is an extension of f . □

Remark 1.2.6. Note that for $f \in \mathbb{R}^k \rightarrow \mathbb{R}$ and $g: \mathbb{R}^n \rightarrow \mathbb{R}^k$

$$\begin{aligned} d(f \circ g(M)) &= f_{x_i}(g(M)) \circ dg^i(M) \\ &= f_{x_i}(g(M)) \circ g_{x_j}^i(M) \circ dM^j \\ &= \nabla f(g(M))^\top Dg(M) \circ dM \end{aligned}$$

under Stratonovich integral.

Remark 1.2.7. If X is a solution of $\text{SDE}(V_1, \dots, V_\ell; Z, X_0)$ up to its explosion time $e(X)$ and $X_0 \in M$, then $X_t \in M$ for $0 \leq t < e(X)$. Moreover, $\text{SDE}(V_1, \dots, V_\ell; Z, X_0)$ is unique up its explosion time.

1.3 Diffusion Process

Let L be a smooth second order elliptic, but not necessarily non degenerate, differential operator on a smooth manifold M , that is,

$$L: C^\infty(M) \rightarrow C^\infty(M)$$

such that on every chart $(U; x^1, \dots, x^d)$

$$Lf(x) = a^{ij}(x) \frac{\partial^2}{\partial x^i \partial x^j} f(x) + b^i(x) \frac{\partial}{\partial x^i} f(x)$$

for some $a^{ij}, b^i \in C^\infty(U)$ and $\sum_{i,j} a^{ij}(x) \xi_i \xi_j \geq 0$.

Remark 1.3.1. If M is equipped with a torsion-free connection ∇ , like a Riemannian manifold, then L can be written as a coordinate-free formulation, that is,

$$Lf := \langle A, \nabla^2 f \rangle + \langle b, \nabla f \rangle,$$

for $b \in \Gamma(TM)$, and A , a symmetric $(2, 0)$ -tensor, where $\langle \cdot, \cdot \rangle$ is the natural dual operation.

For $f \in C^2(M)$ and $\omega \in W(M)$, let

$$M^f(\omega)_t = f(\omega_t) - f(\omega_0) - \int_0^t Lf(\omega_s) ds.$$

Definition 1.3.2. An $\mathbb{F}$ -adapted process X defined on a filtered $(\Omega, \mathbb{F}, \mathbb{P})$ and that is valued in M is called a diffusion process generated by L if X is a M -valued semimartingale w.s.t. $\mathbb{F}$ up to $e(X)$ and

$$M^f(X)_t = f(X_t) - f(X_0) - \int_0^t Lf(X_s) ds, \quad 0 \leq t < e(X),$$

is a $\mathbb{F}$ -local martingale for all $f \in C^\infty(M)$.

A probability measure μ defined on $(W(M), \mathcal{F}_{M,\infty})$ is called a diffusion measure generated by L if

$$M^f(\omega)_t = f(\omega_t) - f(\omega_0) - \int_0^t Lf(\omega_s) ds, \quad 0 \leq t < e(\omega),$$

is a local $\mathbb{F}_M$ local martingale for all $f \in C^\infty(M)$.

Remark 1.3.3. If X is an L -diffusion, then $\mu^X = X_{\#} \mathbb{P}$ is an L -diffusion measure. Conversely, if μ is an L -diffusion measure, then the coordinate process is an L -diffusion process.

For given L , assume locally

$$L = \frac{1}{2} \sum_{i,j=1}^d a^{ij}(x) \frac{\partial^2}{\partial x^i \partial x^j} + \sum_{i=1}^d b^i(x) \frac{\partial}{\partial x^i},$$

where $(a^{ij}(x))$ is positive semi-definite. Define

$$\Gamma(f, g) = L(fg) - fLg - gLf, \quad f, g \in C^\infty(M).$$

So locally

$$\Gamma(f, g) = a^{ij} \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial x^j}.$$

And the semi-elliptic means that for any $f \in C^\infty(M)$, $\Gamma(f, f) \geq 0$. Combining this with bilinearity, it implies that if $g^1, \dots, g^n$, the matrix $(\Gamma(g^i, g^j))$ is positive semimartingale. Here is another explanation for Γ . For any $f, g \in C^\infty(M)$, let $h = fg$. Let $Y = f(X)$ and $Z = g(X)$, so $h(X) = YZ$. First,

$$Y_t Z_t = h(X)_t = Y_0 Z_0 + M_t^h + \int_0^t L(fg)(X_s) ds.$$

On the other hand, because

$$\begin{aligned} Y_t &= Y_0 + M_t^f + \int_0^t Lf(X_s) ds \\ Z_t &= Z_0 + M_t^g + \int_0^t Lg(X_s) ds. \end{aligned}$$

and M^f, M^g are local martingales, by Itô formula,

$$\begin{aligned} Y_t Z_t &= Y_0 Z_0 + \int_0^t f(X_s) dM_s^g + \int_0^t g(X_s) dM_s^f \\ &\quad + \int_0^t (fLg + gLf)(X_s) ds \\ &\quad + \langle M^f, M^g \rangle_t. \end{aligned}$$

Therefore, we have

$$\int_0^t \Gamma(f, g)(X_s) ds - \langle M^f, M^g \rangle_t = \int_0^t f(X_s) dM_s^g + \int_0^t g(X_s) dM_s^f - M_t^h.$$

Note that the LHS is a process with bounded variation and the RHS is a continuous local martingale, which implies that the RHS is constant, i.e., 0. Then we have

$$M_t^h = \int_0^t f(X_s) dM_s^g + \int_0^t g(X_s) dM_s^f.$$

Moreover,

$$\langle Y, Z \rangle_t = \langle M^f, M^g \rangle_t = \int_0^t \Gamma(f, g)(X_s) ds.$$

Existence of Diffusion Process. Assume M is a submanifold of $\mathbb{R}^N$, which is closed. Let $z = (z^1, \dots, z^N)$ be coordinates in $\mathbb{R}^N$ and f^α ($\alpha = 1, 2, \dots, N$) are coordinate maps $M \rightarrow \mathbb{R}^N$ by $f^\alpha(z) = z^\alpha$. Let

$$\tilde{a}^{\alpha\beta} = \Gamma(f^\alpha, f^\beta), \quad \tilde{b}^\alpha = Lf^\alpha.$$

Then they are in $C^\infty(M)$ and $(\tilde{a}^{\alpha\beta})$ is positive semi-definite. By the closedness of M , $\tilde{a}, \tilde{b}$ can be extended to $\mathbb{R}^N$ and let

$$\tilde{L} = \frac{1}{2} \sum_{\alpha, \beta=1}^N \tilde{a}^{\alpha\beta} \frac{\partial^2}{\partial z^\alpha \partial z^\beta} + \sum_{\alpha=1}^N \tilde{b}^\alpha \frac{\partial}{\partial z^\alpha}$$

be defined on $\mathbb{R}^N$. It is a true extension of L .

Lemma 1.3.4. Suppose $f \in C^\infty(M)$. Then for any $\tilde{f} \in C^\infty(\mathbb{R}^N)$ which extends f from M to $\mathbb{R}^N$, $\tilde{L}\tilde{f} = Lf$ on M .

Proof. Let $(x^1, \dots, x^d)$ be a local coordinate on M . Note that

$$f(x) = \tilde{f}(f^1(x), \dots, f^N(x)).$$

Then

$$\begin{aligned} Lf(x) &= L\tilde{f}(f^1(x), \dots, f^N(x)) \\ &= \frac{1}{2}a^{ij}(x) \frac{\partial^2 \tilde{f}(f^1(x), \dots, f^N(x))}{\partial x^i \partial x^j} + b^i(x) \frac{\partial \tilde{f}(f^1(x), \dots, f^N(x))}{\partial x^i} \\ &= \tilde{L}\tilde{f}(z). \end{aligned}$$

□

For $\tilde{L}$ on $\mathbb{R}^n$, if $\tilde{a} = \tilde{\sigma}\tilde{\sigma}^\top$, it is the generator of diffusion process that satisfies

$$X_t = X_0 + \int_0^t \tilde{\sigma}(X_s) dW_s + \int_0^t \tilde{b}(X_s) ds.$$

We can clearly choose $\tilde{\sigma} = \tilde{a}^{\frac{1}{2}}$ and W a standard Brownian motion in $\mathbb{R}^N$ independent of X_0 .

Remark 1.3.5. The existence of solution up to a stopping time needs the local Lipschitz of $\tilde{\sigma}$, which can be guaranteed by the local Lipschitz of $\tilde{a}$.

Moreover, for such diffusion process X on $\mathbb{R}^N$, $\mu^X = X_\# \mathbb{P}$, the distribution on $W(\mathbb{R}^N)$, is concentrated on $W(M)$, i.e., X is in fact a L -diffusion on M . In fact, we have the following theorem.

Theorem 1.3.6. *If L is a smooth second order semi-elliptic operator on a C^∞ -manifold M and μ_0 is a probability measure on M , then there exists a unique L -diffusion measure with initial distribution μ_0 .*

Remark 1.3.7. The existence and uniqueness of such $\mathbb{P}_{\mu_0}$ guarantee that the L martingale problem is well-posed. So if X is a L -diffusion process, then it is a strong Markov process w.s.t. $\mathbb{F}^X$ and its transition group $P_t f = \mathbb{E}^x[X_t]$ has the generator extending L . In this case, in fact, X is a strong Markov process w.s.t. $\mathbb{F}$.

Driving by Brownian Motion. Let M be a Riemannian manifold and consider all solutions of SDE

$$dX_t = V_\alpha(X_t) \circ dW_t^\alpha + V_0(X_t)dt,$$

where $V_\alpha, V_0 \in \Gamma(TM)$ and $W = (W^\alpha)$ is an ℓ -dimensional Brownian motion. First, by Itô formula,

$$dV_\alpha f(X_t) = V_0(V_\alpha f(X_t))dt + V_\beta(V_\alpha f(X_t)) \circ dW_t^\beta,$$

which implies that

$$d\langle V_\alpha f(X_t), W_t^\alpha \rangle = V_\beta(V_\alpha f(X_t))d\langle W^\beta, W^\alpha \rangle_t = \sum_\alpha V_\alpha(V_\alpha f(X_t))dt$$

Therefore, we have

$$\begin{aligned} f(X_t) - f(X_0) &= \int_0^t V_0 f(X_t)dt + \int_0^t V_\alpha f(X_t) \circ dW_t^\alpha \\ &= \int_0^t V_0 f(X_t)dt + \int_0^t V_\alpha f(X_t) dW_t^\alpha + \frac{1}{2} \langle V_\alpha f(X_t), W^\alpha \rangle_t \end{aligned}$$

$$= \int_0^t V_\alpha f(X_t) dW_t^\alpha + \int_0^t \left(V_0 f(X_t) + \frac{1}{2} \sum_\alpha V_\alpha (V_\alpha f(X_t)) \right) dt$$

Denote $V_\alpha(V_\alpha f(x)) = V_\alpha^2 f(x)$ and let

$$L = \frac{1}{2} \sum_\alpha V_\alpha^2 + V_0,$$

which is obviously a semi-elliptic second order differential operator. From above,

$$f(X_t) - f(X_0) - \int_0^t Lf(X_t) dt = \int_0^t V_\alpha f(X_t) dW_t^\alpha.$$

So a solution to $dX_t = V_\alpha(X_t) \circ dW_t^\alpha + V_0(X_t)dt$ is a diffusion process generated by L . Moreover, assume that V_α, V_0 satisfy

$$\sup_{t \in [0, T]} \sup_{x \in U} \mathbb{E} [|D^\alpha f(X_t)| \mid X_0 = x] < \infty$$

for all $f \in C_c^\infty(M)$, $T > 0$, coordinate chart U with $\bar{U}$ compact, and index α , where D^α is the usual derivative.

Remark 1.3.8. In fact, assume $M \subset \mathbb{R}^n$ and V_α, V_0 are the restriction of $\mathbb{R}^n$ -vector fields $\tilde{V}_\alpha, \tilde{V}_0$ on M . If $\tilde{V}_\alpha = (a_\alpha^j)$, $\tilde{V}_0 = (b^j)$ satisfy the condition of the existence and uniqueness of solution of SDE, then it satisfies the above condition.

Then define

$$u(t, x) = \mathbb{E}[f(X_t) \mid X_0 = x], \quad f \in C_c^\infty(M).$$

Because $f \in C_c^\infty(M)$,

$$M_t^f = f(X_t) - f(X_0) - \int_0^t Lf(X_s) ds$$

is a martingale, which implies that $\mathbb{E}[M_t^f \mid X_0] = M_0^f = 0$. Therefore,

$$u(t, x) = \mathbb{E}[f(X_t) \mid X_0 = x] = f(x) + \int_0^t \mathbb{E}[Lf(X_s) \mid X_0 = x] ds,$$

called the Dynkin's formula. Moreover, for $u_t = u(t, \cdot)$,

$$\begin{cases} \frac{\partial u_t}{\partial t} = Lu_t, \\ u(0, x) = f(x). \end{cases}$$

This result can be extended to so-called Feynman-Kac formula. Let

$$v(t, x) = \mathbb{E} \left[\exp \left(\int_0^t g(X_s) ds \right) f(X_t) \mid X_0 = x \right].$$

Then $v_t = v(t, \cdot)$ satisfies

$$\begin{cases} \frac{\partial v_t(x)}{\partial t} = Lv_t(x) + g(x)v_t(x), \\ v(0, x) = f(x). \end{cases}$$

Chapter 2

Stochastic Differential Geometry

2.1 Horizontal Lift

Let M be d -dimensional smooth manifold equipped with an affine connection ∇ . Let Γ_{ij}^k be the corresponding Christoffel symbol, i.e.,

$$\nabla_{X_i} X_j = \Gamma_{ij}^k X_k,$$

where $X_i = \frac{\partial}{\partial x^i}$. ∇ can induce concepts, such as, parallel moving ($\nabla_{\dot{c}(t)} V = 0$) and geodesic ($\nabla_{\dot{c}(t)} \dot{c}(t) = 0$).

For $x \in M$, a frame at x is a $\mathbb{R}$ -linear isomorphism $u: \mathbb{R}^d \rightarrow T_x M$. Let $F(M)_x$ be the set of all frameworks at x . $GL(d, \mathbb{R})$ acts on $F(M)_x$ as $u \mapsto ug = u \circ g$. Let $F(M) = \bigcup_{x \in M} F(M)_x$, called the frame bundle of dimension $d + d^2$, with the canonical projection $\pi: F(M) \rightarrow M$. Note that each $F(M)_x \cong GL(d, \mathbb{R})$. In fact, $(F(M), M, GL(d, \mathbb{R}))$ is a principal bundle $M \cong F(M)/GL(d, \mathbb{R})$. Let

$$F(M) \times_{GL(d, \mathbb{R})} \mathbb{R}^d := F(M) \times \mathbb{R}^d / \sim,$$

where $(ug, v) \sim (u, gv)$. Then define

$$\Phi: F(M) \times_{GL(d, \mathbb{R})} \mathbb{R}^d \rightarrow TM$$

by

$$\Phi([u, v]) = u(v) \in T_{\pi(u)} M.$$

Φ is a linear isomorphism and so $F(M) \times_{GL(d, \mathbb{R})} \mathbb{R}^d \cong TM$.

Consider the canonical projection $\pi: F(M) \rightarrow M$, because it is a smooth submersion and $\pi^{-1}(x) = F(M)_x \cong GL(d, \mathbb{R})$, for

$$d\pi_u: T_u F(M) \rightarrow T_{\pi(u)} M,$$

the vertical space

$$V_u F(M) := \ker(d\pi_u) = T_u F(M)_{\pi(u)} \cong \mathfrak{g}(d, \mathbb{R}).$$

Let M be equipped with an affine connection ∇ . Let $c(t)$ be a smooth curve on M and $u(t)$ be a lift of $c(t)$ on $F(M)$, i.e., $\pi(u(t)) = c(t)$. Because $u(t) \in F(M)_{c(t)}$, for any $e \in \mathbb{R}^d$, $u_e(t) = u(t)e \in T_{c(t)} M$, which means $u_e \in \Gamma(TM)$ is a vector field along $c(t)$. We say $u(t)$ is a horizontal lift if for any $e \in \mathbb{R}^d$, u_e is parallel along c , i.e.,

$$\nabla_{\dot{c}(t)} u_e(t) \equiv 0.$$

Lemma 2.1.1. *For any smooth curve $c: (-\varepsilon, \varepsilon) \rightarrow M$, let $u_0 \in F(M)_{c(0)}$. Then there exists a unique $u: (-\varepsilon, \varepsilon) \rightarrow F(M)$ such that*

$$\pi \circ u = c, \quad u(0) = u_0, \quad \nabla_{\dot{c}(t)} \dot{u}_a(t) = 0, \quad \forall a \in \mathbb{R}^d.$$

Moreover, u depends smoothly on (c, u_0) , and if $u_1 = u_0 g$, the $t \mapsto u(t)g$ is the unique horizontal lift starting from u_1 .

Proof. WTLG, assume $c((-\varepsilon, \varepsilon))$ is contained in a chart $(U; x^1, \dots, x^d)$ and let Γ_{ij}^k be the corresponding Christoffel symbol for ∇ . Let $e_1, \dots, e_d$ be the canonical basis of $\mathbb{R}^d$. Then a vector field $V(t) = V^i(t) \frac{\partial}{\partial x^i} \Big|_{c(t)}$ along $c(t)$ is parallel to $c(t)$ if and only if $V(t)$ satisfies

$$\dot{V}^i(t) + A_k^i(t) V^k(t) = 0, \quad \forall i,$$

where $A_k^i(t) = \Gamma_{jk}^i(c(t)) \dot{c}^j(t)$. For each e_j , above equation has a unique solution

$$V_{e_j}(t) = V_j^i(t) \frac{\partial}{\partial x^i} \Big|_{c(t)}$$

with initial value $V_{e_j}(0) = u_0 e_j = u_{0,j}^i \frac{\partial}{\partial x^i} \Big|_{c(0)}$. So the matrix $U(t) = (V_j^i(t))$ is the unique solution of ODE

$$\dot{U}(t) + A(t)U(t) = 0, \quad U(0) = U_0 := (u_{0,j}^i).$$

Moreover, because

$$\frac{d}{dt} \det U(t) = \text{tr} \left(U(t)^{-1} \dot{U}(t) \right) \det U(t) = -\text{tr}(A(t)) \det U(t),$$

and $u_0 \in F(M)_{c(0)}$ (i.e., $\det u_0 \neq 0$),

$$\det U(t) = \det U_0 \cdot \exp \left(- \int_{t_0}^t \text{tr}(A(s)) ds \right) \neq 0,$$

it means $U(t) \in GL(d, \mathbb{R})$ for all t . Let

$$u(t)a := V_j^i(t) a^j \frac{\partial}{\partial x^i} \Big|_{c(t)}$$

for any $a = (a^j) \in \mathbb{R}^d$. Then $u(t) \in F(M)_{c(t)}$ and satisfies $u(0) = u_0$ and the parallel condition. The uniqueness of $u(t)$ is directly obtained by the uniqueness of solution of the ODE, which also implies that $t \mapsto u(t)g$ is the unique lift starting from u_1 . Also, the smoothness on initial values is also by the theory of ODE. $\square$

Remark 2.1.2. Moreover, give a curve $c(t)$ on M , let $u(t), v(t)$ be its horizontal lifts starting from u_0, v_0 respectively, because $u_0 = v_0(v_0^{-1}u_0)$, by above

$$u(t) = v(t)(v_0^{-1}u_0).$$

Moreover, for any t_0, t_1 , it is obvious that

$$\mathcal{P}_{c, t_0, t_1} = u_{t_1} u_{t_0}^{-1}: T_{c(t_0)} M \rightarrow T_{c(t_1)} M$$

is the parallel moving along $c(t)$ by considering $V(t) = u_t(u_{t_0}^{-1}(X))$ and the uniqueness of parallel moving.

Remark 2.1.3. Moreover, if $\dot{c}(0) = X = X^i \frac{\partial}{\partial x^i} \Big|_{c(0)}$, then we can see

$$\dot{u}(0) = -\Gamma_{\ell k}^i(c(0))X^\ell u_{0,j}^k \frac{\partial}{\partial v_j^i} \Big|_{u_0} + X^i \frac{\partial}{\partial x^i} \Big|_{u_0}$$

which implies $c(0)$, $\dot{c}(0)$, and u_0 uniquely determined $\dot{u}(0) \in T_{u_0}F(M)$.

For a given $X \in T_x M$, let c be a smooth curve in M with $c(0) = x$ and $\dot{c}(0) = X$. Let $u \in F(M)_x$. Let $u(t)$ be the unique horizontal lift of c starting from u . Then define

$$X_u^H = \dot{u}(0) \in T_u F(M),$$

which is well-defined because it is independent of the choice of c , called a horizontal vector. Let

$$H_u F(M) := \{X_u^H : X \in T_x M\} \subset T_u F(M),$$

called the horizontal space.

Lemma 2.1.4. *For the canonical projection $\pi: F(M) \rightarrow M$ and any $u \in F(M)$ with $x = \pi(u)$,*

$$d\pi_u: H_u F(M) \rightarrow T_x M$$

is an isomorphism. Therefore,

$$T_u F(M) = V_u F(M) \oplus H_u F(M).$$

Proof. For any $X_u^H \in H_u F(M)$, $X_u^H = \dot{u}(0)$ for some $u(t)$ which is a horizontal lift of $c(t)$ with $c(0) = x$ and $\dot{c}(0) = X$ and starting from u with $\pi(u) = x$. Then

$$d\pi_u(X_u^H) = \frac{d}{dt} \Big|_{t=0} \pi(u(t)) = \frac{d}{dt} \Big|_{t=0} c(t) = X.$$

And from the construction, we have seen $X \neq 0$ implies that $X_u^H \neq 0$. □

For any $e \in \mathbb{R}^d$ and $u \in F(M)$, $ue \in T_{\pi(u)} M$ and so we can consider

$$H_e(u) := (ue)_u^H = \dot{u}(0) \in H_u F(M),$$

where $u(t)$ is the horizontal lift of $c(t)$ for $c(0) = \pi(u)$ and $\dot{c}(0) = ue$ starting from u . By above, we know

$$dg_u H_e(u) = H_{g^{-1}e}(ug)$$

where $g: F(M) \rightarrow F(M)$ is $g(u) = u \circ g = ug$, because

$$dg_u H_e(u) = \frac{d}{dt} \Big|_{t=0} u(t)g.$$

Remark 2.1.5. Note that because

$$(d\pi_u)^{-1}: X \mapsto X_u^H$$

is $\mathbb{R}$ -linear,

$$H_{\lambda e + \mu e'}(u) = \lambda H_e(u) + \mu H_{e'}(u), \quad \forall e, e' \in \mathbb{R}^d, u \in F(M), \lambda, \mu \in \mathbb{R}.$$

Let $e_1, \dots, e_d$ be the canonical basis of $\mathbb{R}^d$. Then

$$H_j(u) = H_{e_j}(u) \in H_u F(M)$$

forms a basis of $H_u F(M)$ for $j = 1, \dots, d$, because for any $v = (v^i) \in \mathbb{R}^d$, i.e., $v = v^i e_i$, then

$$H_v(u) = \sum_{i=1}^d v^i H_i(u).$$

$\{H_i\}_{i=1}^d$ are called the fundamental horizontal vector fields.

Remark 2.1.6. For any $X \in \Gamma(TM)$, its unique horizontal lift vector field $X^H \in \Gamma(TF(M))$ is defined as

$$X^H(u) := (X(\pi(u)))_u^H \in T_u F(M), \quad \forall u \in F(M).$$

Moreover, by definition, $d\pi_u(X^H(u)) = X(\pi(u))$, i.e.,

$$\pi_* X^H = X.$$

This is the reason why X^H is called a lift.

Let $c(t)$ be a smooth curve on M and $u(t)$ be the horizontal lift of it starting from u_0 . Then we can consider a curve $w(t)$ on $\mathbb{R}^d$, called the anti-development of $c(t)$ w.s.t. u_0 , defined as

$$w_{u_0}(t) = \int_0^t u(s)^{-1} \dot{c}(s) ds,$$

which is because $\dot{c}(t) \in T_{c(t)}M$ and $u(t)^{-1}: T_{c(t)}M \rightarrow \mathbb{R}^d$. But note that $w_{u_0}(t)$ depends on the choice of u_0 . Let $v(t)$ be the horizontal lift of $c(t)$ starting from v_0 . If $u_0 = v_0 g$, i.e., $v_0 = u_0 g^{-1}$, then $v(t) = u(t) g^{-1}$ and thus

$$w_{v_0}(t) = \int_0^t v(s)^{-1} \dot{c}(s) ds = g \int_0^t u(s)^{-1} \dot{c}(s) ds = g w_{u_0}(t).$$

Fix u_0 and let $w(t) = w_{u_0}(t)$. Because $\dot{w}(t) = u(t)^{-1} \dot{c}(t)$, i.e., $u(t) \dot{w}(t) = \dot{c}(t)$,

$$H_{\dot{w}(t)}(u(t)) = (u(t) \dot{w}(t))_{u(t)}^H = (\dot{c}(t))_{u(t)}^H = \dot{u}(t),$$

i.e.

$$\dot{u}(t) = \dot{w}^i(t) H_i(u(t)).$$

Note that from the definition of $H_i(u(t))$ this is linear in $u(t)$, when given $w(t)$ and u_0 , this ODE has a unique solution $u(t)$ called the development of $w(t)$ in $F(M)$ and its projection $c(t) = \pi(u(t))$ is called the development of $w(t)$ in M .

Orthogonal Frame. When considering (M, g) be a Riemannian manifold with the Levi-Civita connection ∇ . Instead of considering $F(M)$, we consider the orthogonal frame $O(M)$, that is, $O(M)_x$ consists of all $u: \mathbb{R}^d \rightarrow T_x M$ orthogonal map. The problem is if we can consider the horizontal lift on $O(M)$. To do that, let

$$u(t)a = V_j^i(t) a^j \frac{\partial}{\partial x^i}$$

be the parallel vector field by solving the ODE

$$\dot{U}(t) + A(t)U(t) = 0.$$

The problem is whether $u(t) \in O(M)_{c(t)}$, that is, whether

$$\langle u(t)e_i, u(t)e_j \rangle = \left\langle V_i^\alpha \frac{\partial}{\partial x^\alpha}, V_j^\beta \frac{\partial}{\partial x^\beta} \right\rangle = V_i^\alpha g_{\alpha\beta} V_j^\beta = (U(t)^\top G(t) U(t))_{ij} = \delta_{ij},$$

i.e., $U(t)^\top G(t) U(t) \equiv I$, where $U(t) = (V_j^i(t))$ and $G(t) = (g_{ij}(c(t)))$, the local expression of metric. Note that, by the consistency of metric

$$\dot{g}_{ij}(c(t)) = \dot{c}(t) \left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle$$

$$\begin{aligned}
&= \left\langle \nabla^{\dot{c}(t)} \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle + \left\langle \frac{\partial}{\partial x^i}, \nabla^{\dot{c}(t)} \frac{\partial}{\partial x^j} \right\rangle \\
&= \dot{c}^\alpha \Gamma_{\alpha i}^k g_{kj} + \dot{c}^\alpha \Gamma_{\alpha j}^k g_{ik} \\
&= A^\top G + GA,
\end{aligned}$$

where $A = (A_k^i)$ for $A_k^i(t) = \Gamma_{jk}^i(c(t))\dot{c}^j(t)$. Moreover, because

$$\dot{U}(t) + A(t)U(t) = 0,$$

we have

$$\begin{aligned}
\frac{d}{dt} U^\top G U &= \dot{U}^\top G U + U^\top \dot{G} U + U^\top G \dot{U} \\
&= -U^\top A^\top G U + U^\top (A^\top G + GA) U - U^\top G (AU) = 0.
\end{aligned}$$

Therefore,

$$U^\top G U \equiv U^\top(0)G(0)U(0) = I,$$

when $u(0) \in O(M)_{c(0)}$. Therefore, the horizontal lift lies on $O(M)$. For Riemannian manifolds, the following $F(M)$ can be replaced by $O(M)$.

2.2 Tensor Fields Scalarization

Let θ be a (r, s) -tensor field on M . For any $u \in F(M)$, let

$$X_i = u e_i \in T_{\pi(u)} M$$

for $i = 1, \dots, d$, where $\{e_i\}$ is the canonical basis of $\mathbb{R}^d$. Then $\{X_i\}$ is a basis of $T_{\pi(u)} M$. Choose $\{X^i\}$ be the its dual basis of $T_{\pi(u)}^* M$. Then under these basis, we have

$$\theta(\pi(u)) = \theta_{j_1 \dots j_s}^{i_1 \dots i_r} X_{i_1} \otimes \dots \otimes X_{i_r} \otimes X^{j_1} \otimes \dots \otimes X^{j_s}.$$

and we define the scalarization

$$\tilde{\theta}: F(M) \rightarrow (\mathbb{R}^d)^{\otimes r} \otimes (\mathbb{R}^d)^{* \otimes s} =: \otimes^{(r,s)} \mathbb{R}^d,$$

as

$$\tilde{\theta}(u) = \theta_{j_1 \dots j_s}^{i_1 \dots i_r} e_{i_1} \otimes \dots \otimes e_{i_r} \otimes e^{j_1} \otimes \dots \otimes e^{j_s},$$

where $\{e^i\}$ is the dual basis of $\{e_i\}$ in $(\mathbb{R}^d)^*$. Note that $\tilde{\theta}$ is $GL(d, \mathbb{R})$ -equivariant, i.e.,

$$\tilde{\theta}(ug) = g\tilde{\theta}(u).$$

If θ is $(0, 0)$ -tensor, i.e., $\theta = f: M \rightarrow \mathbb{R}$,

$$\tilde{f} = f \circ \pi: F(M) \rightarrow \mathbb{R}.$$

Therefore, $\tilde{\theta}$ can be considering as a lift of θ on $F(M)$.

Remark 2.2.1. For a finite dimensional vector space V , let $\{v_i\}$ be a basis of V and $\{v_i^*\}$ be its dual basis in V^* . If

$$w_j = g_j^i v_i$$

and $\{w_j\}$ is another basis, i.e., $G = (g_j^i)_{i \times j} \in GL$, then for $w_\ell^* = \sum_k h_k^\ell (v_k)^*$, we have the matrix

$$H = (h_k^\ell)_{\ell \times k} = G^{-1}.$$

Moreover, for $u \in F(M)$, $u^{-1}: T_{\pi(u)}M \rightarrow \mathbb{R}^d$ and so we can consider $u^*: T_{\pi(u)}^*M \rightarrow (\mathbb{R}^d)^*$, which follows that we can defined $u^{-1}: \otimes^{(r,s)} T_{\pi(u)}M \rightarrow \otimes^{(r,s)} \mathbb{R}^d$ as

$$u^{-1} = \underbrace{u^{-1} \otimes \cdots \otimes u^{-1}}_r \times \underbrace{u^* \otimes \cdots \otimes u^*}_s.$$

Then it is obvious that for $\theta \in \Gamma(\otimes^{(r,s)} TM)$

$$\tilde{\theta}(u) = u^{-1}\theta(\pi(u)).$$

On the other hand, for $u: \mathbb{R}^d \rightarrow T_x M$, consider $(u^{-1})^*: (\mathbb{R}^d)^* \rightarrow T_{\pi(u)}^*M$. Then it can extend u to be defined on $\otimes^{(r,s)} (\mathbb{R}^d)^* \rightarrow \otimes^{(r,s)} T_{\pi(u)}^*M$ as

$$u = \underbrace{(u^{-1})^* \otimes \cdots \otimes (u^{-1})^*}_r \otimes \underbrace{u \otimes \cdots \otimes u}_s,$$

where $\otimes^{(r,s)} (\mathbb{R}^d)^* = \otimes^r (\mathbb{R}^d)^* \otimes \otimes^s \mathbb{R}^d$ and $\otimes^{(r,s)} T_{\pi(u)}^*M = \otimes^r T_{\pi(u)}^*M \otimes \otimes^s T_{\pi(u)}M$. Then for any $\theta \in \Gamma(\otimes^{(r,s)} TM)$,

$$\theta_{\pi(u)}: \otimes^{(r,s)} T_{\pi(u)}^*M \rightarrow \mathbb{R}, \quad \tilde{\theta}_u: \otimes^{(r,s)} (\mathbb{R}^d)^* \rightarrow \mathbb{R},$$

they have

$$\theta_{\pi(u)} \circ u = \tilde{\theta}(u),$$

because $\{(u^{-1})^* e^j\}$ is the dual basis of $\{ue_i\}$ in $T_{\pi(u)}^*M$. For $f \in C^\infty(M)$, clearly $f_{\pi(u)} = \tilde{f}(u)$. And so

$$\nabla_X f(\pi(u)) = \widetilde{\nabla_X f}(u).$$

Proposition 2.2.2. *Let $X \in \Gamma(TM)$ and $\theta \in \Gamma(\otimes^{(r,s)} TM)$. Then*

$$\widetilde{\nabla_X \theta} = X^H \tilde{\theta},$$

i.e., $\widetilde{\nabla_X \theta}(u) = X_u^H \tilde{\theta}$ for all $u \in F(M)$, where $X_u^H \tilde{\theta}$ acts on each component since $X_u^H \in T_u F(M)$.

Proof. Fix $u \in F(M)$. Let $c(t)$ be a curve in M with $c(0) = \pi(u)$ and $\dot{c}(0) = X_{\pi(u)}$. Let $u(t)$ be the unique horizontal lift of $c(t)$ starting from $u(0) = u$. Then as we know, $\tau_t = u(t)u^{-1}$ is the parallel moving along $c(t)$. So we have

$$\left. \frac{d}{dt} \right|_{t=0} \tau_t^{-1} \theta(c(t)) = \nabla_X \theta(\pi(u)).$$

On the other hand, because $\pi(u(t)) = c(t)$, we have

$$\tilde{\theta}(u(t)) = u(t)^{-1} \theta(c(t)).$$

Moreover, because $X_u^H = \dot{u}(0)$,

$$\begin{aligned} X_u^H \tilde{\theta} &= \left. \frac{d}{dt} \right|_{t=0} \tilde{\theta}(u(t)) \\ &= \left. \frac{d}{dt} \right|_{t=0} u(t)^{-1} \theta(c(t)) \\ &= u^{-1} \left. \frac{d}{dt} \right|_{t=0} \tau_t^{-1} \theta(c(t)) \\ &= u^{-1} (\nabla_X \theta)(\pi(u)) = \widetilde{\nabla_X \theta}(u), \end{aligned}$$

where the first equality is because $\tilde{\theta}$ is a tensor field on $\mathbb{R}^d$, i.e. the coordinate does not change along time. $\square$

Remark 2.2.3. By the definition of $X_u^H \tilde{\theta}$, because it is a tensor field on $\mathbb{R}^d$,

$$(X_u^H \tilde{\theta})(v^1, \dots, v^r, w_1, \dots, w_s) = X_u^H \left(\tilde{\theta}(u)(v^1, \dots, v^r, w_1, \dots, w_s) \right)$$

for any $v^i \in (\mathbb{R}^d)^*$ and $w_j \in \mathbb{R}^d$.

Remark 2.2.4. Above is also true for $f \in C^\infty(M)$.

$$\begin{aligned} X_u^H \tilde{f} &= \left. \frac{d}{dt} \right|_{t=0} \tilde{f}(u(t)) \\ &= \left. \frac{d}{dt} \right|_{t=0} f(\pi(u(t))) = \left. \frac{d}{dt} \right|_{t=0} f(c(t)) \\ &= \nabla_X f(c(0)) = \nabla_X f(\pi(u)) = \widetilde{\nabla_X f}(u). \end{aligned}$$

Example 2.2.5 (Hessian). For any $f \in C^\infty(M)$, consider the Hessian $\nabla^2 f \in \Gamma(\otimes^{0,2} TM)$,

$$\begin{aligned} \nabla^2 f_{\pi(u)}(ue_i, ue_j) &= (\nabla_{ue_j} \nabla f)_{\pi(u)}(ue_i) = (\widetilde{\nabla_{ue_j} \nabla f})(u)(e_i) \\ &= \left((ue_j)_u^H \widetilde{\nabla f} \right)(e_i) = H_j \left(\widetilde{\nabla f}(u)(e_i) \right) \\ &= H_j \left(\nabla f_{\pi(u)}(ue_i) \right) = H_j \left((\nabla_{ue_i} f)_{\pi(u)} \right) \\ &= H_j \left(\widetilde{\nabla_{ue_i} f}(u) \right) = H_j H_i \tilde{f} \end{aligned}$$

where $\tilde{f} = f \circ \pi$.

2.3 Stochastic Horizontal Lift

Given a SDE on $F(M)$:

$$dU_t = H(U_t) \circ dW_t = \sum_{i=1}^d H_i(U_t) \circ dW_t^i, \quad (2.1)$$

where W is an $\mathbb{R}^d$ -valued semimartingale and $\{H_i\}_{i=1}^d$ are the fundamental horizontal vector fields on $F(M)$.

Definition 2.3.1. (1) An $F(M)$ -valued semimartingale is said to be horizontal if there exists an $\mathbb{R}^d$ -valued semimartingale W with $W_0 = 0$ such that SDE (2.1) holds. Then W is called the anti-development of U (or of its projection $X = \pi(U)$).

(2) Let W be an $\mathbb{R}^d$ -valued semimartingale with $W_0 = 0$ and U_0 is an $F(M)$ -valued, $\mathcal{F}_0$ -measurable random variable. The solution of SDE (2.1) with initial condition U_0 is called a stochastic development of W in $F(M)$, so is its projection $X = \pi(U)$.

(3) Let X be an M -valued semimartingale. An $F(M)$ -valued horizontal semimartingale U such that $\pi(U) = X$ is called a stochastic horizontal lift of X .

Note that $W \mapsto U$ is by considering the solution of SDE (2.1), and $U \mapsto X$ is just by projection, but we need $X \mapsto U$ and $U \mapsto W$.

Assume $M \subset \mathbb{R}^N$ is a submanifold that is closed. Let $f = (f^\alpha): M \rightarrow \mathbb{R}^N$ be the coordinate function and let lift it to

$$\tilde{f}: F(M) \rightarrow M \subset \mathbb{R}^N$$

as $\tilde{f} = f \circ \pi$, i.e. $\tilde{f}(u) = f(\pi(u)) = \pi(u)$ as in $\mathbb{R}^N$.

For any M -valued semimartingale, we regard $X = (X^\alpha)_{\alpha=1}^N = (f^\alpha(X))_{\alpha=1}^N \in \mathbb{R}^N$ as an $\mathbb{R}^N$ -valued semimartingale.

For any $x \in M$, let $P(x): \mathbb{R}^N \rightarrow T_x M$ be the orthogonal projection by viewing $T_x M \subset \mathbb{R}^N$. Let $P_\alpha(x) = P(x)e_\alpha \in T_x M$ and so $P_\alpha \in \Gamma(TM)$.

Remark 2.3.2. For the smoothness of P_α , first we can equip M with the induced metric. Then on each local chart, there exists a smooth orthonormal frames E_i on TM , which implies that

$$P_\alpha = \sum_i \langle e_\alpha, E_i \rangle E_i.$$

So P_α is smooth.

Define $P_\alpha^H \in \Gamma(TF(M))$ as

$$P_\alpha^H(u) = (P_\alpha(\pi(u)))_u^H \in T_u F(M), \quad \forall u \in F(M).$$

By using the fundamental horizontal lifts,

$$P_\alpha^H(u) = (u(u^{-1}P_\alpha(\pi(u))))_u^H = (u^{-1}P_\alpha(\pi(u)))^i H_i(u).$$

Lemma 2.3.3. *For notations as above, when viewing $T_x M \subset \mathbb{R}^N$, we have*

$$P_\alpha^H(u)\tilde{f} = P_\alpha(\pi(u)), \quad P_\alpha(\pi(u))H_i(u)\tilde{f}^\alpha = ue_i.$$

Proof. Let $u(t)$ be the horizontal lift from $u(0) = u$ of a curve $c(t)$ on M with $c(0) = \pi(u)$ and $\dot{c}(0) = \tilde{f}(u) = P_\alpha(\pi(u))$. It follows that $P_\alpha^H(u) = \dot{u}(0)$. Because $\tilde{f}: F(M) \rightarrow \mathbb{R}^N$ and $P_\alpha^H(u) \in T_u F(M)$,

$$P_\alpha^H(u)\tilde{f} = \frac{d}{dt} \Big|_{t=0} \tilde{f}(u(t)) = \frac{d}{dt} \Big|_{t=0} \pi(u(t)) = \frac{d}{dt} \Big|_{t=0} c(t) = P_\alpha(\pi(u)).$$

Next, let $v(t)$ be the horizontal lift start from $v(0) = u$ of curve $x(t)$ with $x(0) = \pi(u)$ and $\dot{x}(0) = ue_i$. So

$$H_i(u)\tilde{f} = \frac{d}{dt} \Big|_{t=0} \tilde{f}(v(t)) = \frac{d}{dt} \Big|_{t=0} \pi(v(t)) = \frac{d}{dt} \Big|_{t=0} x(t) = ue_i,$$

which implies that $H_i(u)\tilde{f} \in T_{\pi(u)} M$ and so

$$P(\pi(u))H_i(u)\tilde{f} = H_i(u)\tilde{f} \Rightarrow P_\alpha(\pi(u))H_i(u)\tilde{f}^\alpha = H_i(u)\tilde{f} = ue_i. \quad \square$$

Remark 2.3.4. For a vector field $H \in \Gamma(TF(M))$, we usually denote $Hg(u) := H(u)g$ for any $u \in F(M)$ and $g: F(M) \rightarrow \mathbb{R}^n$. Under this notation,

$$P_\alpha^H \tilde{f}(u) = P_\alpha(\pi(u)), \quad P_\alpha(\pi(u))H_i \tilde{f}^\alpha(u) = ue_i.$$

Remark 2.3.5. Note that above two identities view $P_\alpha(\pi(u))$ as a vector in $\mathbb{R}^N$. More precisely, for any $g \in C^\infty(M)$, $g \circ \pi \in C^\infty(F(M))$. We similarly have

$$H_i(u)(g \circ \pi) = \frac{d}{dt} \Big|_{t=0} g(c(t)) = (ue_i)g.$$

and

$$P_\alpha^H(u)(g \circ \pi) = P_\alpha(\pi(u))g$$

or in other words,

$$d\pi_u(P_\alpha^H(u)) = P_\alpha(\pi(u)),$$

which is the definition of horizontal lift.

Lemma 2.3.6. Assume $M \subset \mathbb{R}^N$ is a submanifold that is closed. If X is an M -valued semimartingale, then

$$X_t = X_0 + \int_0^t P(X_s) \circ dX_s = X_0 + \int_0^t P_\alpha(X_s) \circ dX_s^\alpha.$$

Proof. Let $Q_\alpha(x) = e_\alpha - P_\alpha(x)$, i.e., $Q_\alpha(x)$ normal to $T_x M$ and $Q_\alpha \in \Gamma(NM)$. Let

$$Y_t = X_0 + \int_0^t P_\alpha(X_s) \circ dX_t^\alpha$$

By the closedness of M , there exists $f \in C^\infty(\mathbb{R}^N)$ such that $f(x) \geq 0$ and $f^{-1}(0) = M$. Then by Itô formula,

$$f(Y_t) = f(X_0) + \int_0^t P_\alpha f(X_t) \circ dX_t^\alpha.$$

Because $P_\alpha \in \Gamma(TM)$ and $f|_M = 0$, $P_\alpha f(x) = 0$ for all $x \in M$. Since X is M -valued, $f(Y_t) = 0$, i.e., $Y \in f^{-1}(0) = M$, and Y is an M -valued semimartingale.

Next, consider a tubular neighborhood U of M and defined the natural projection $\tilde{h}: U \rightarrow M$, which can be extended a smooth $h: \mathbb{R}^N \rightarrow M$. Because $Q_\alpha \in \Gamma(NM)$, by choosing Fermi coordinates, we have

$$Q_\alpha h(x) = 0 \Rightarrow P_\alpha h(x) = \frac{\partial h}{\partial x^\alpha}.$$

So

$$\begin{aligned} Y_t &= h(Y_t) = h(X_0) + \int_0^t P_\alpha h(X_s) \circ dX_s^\alpha \\ &= X_0 + \int_0^t \frac{\partial h}{\partial x^\alpha} \circ dX_s^\alpha \\ &= h(X_t) = X_t. \end{aligned}$$

□

Theorem 2.3.7. A horizontal semimartingale U on $F(M)$ has a unique anti-development W . In fact,

$$W_t = \int_0^t U_s^{-1} P_\alpha(X_s) \circ dX_s^\alpha,$$

for $X = \pi(U)$.

Proof. By definition, W should be

$$dU_t = H_i(U_t) \circ dW_t^i.$$

Let $\tilde{f}$ be defined as above. Then

$$\tilde{f}(U_t) = f(\pi(U_t)) = f(X_t) = X_t \in M \subset \mathbb{R}^N.$$

Then X_t should satisfy

$$dX_t^\alpha = H_i \tilde{f}^\alpha(U_t) \circ dW_t^i.$$

Therefore, by above formulas, we have

$$U_t^{-1} P_\alpha(X_t) \circ dX_t^\alpha = U_t^{-1} P_\alpha(X_t) H_i \tilde{f}^\alpha(U_t) \circ dW_t^i = e_i dW_t^i = dW_t$$

By the uniqueness of SDE, W_t uniquely exists and

$$W_t = \int_0^t U_s^{-1} P_\alpha(X_s) \circ dX_s^\alpha.$$

□

Remark 2.3.8. Note that if U_t^1 and U_t^2 are two horizontal lifts of X starting from U_0^1 and $U_0^2 = U_0^1 g$, then by above, we obviously have

$$W_t^1 = g^{-1} W_t^2.$$

Remark 2.3.9. On the other hand,

$$\begin{aligned} P_\alpha(X_t) \circ dX_t^\alpha &= P_\alpha(X_t) H_i \widetilde{f}^\alpha(U_t) \circ dW_t^i \\ &= U_t e_i \circ dW_t^i. \end{aligned}$$

Then by above,

$$X_t = X_0 + \int_0^t U_s e_i \circ dW_s^i.$$

In formally, we just denote

$$X_t = X_0 + \int_0^t U_s \circ dW_s.$$

Similarly, because $dX_t = P_\alpha(X_t) \circ dX_t^\alpha$, we can informally write

$$W_t = \int_0^t U_s^{-1} \circ dX_s.$$

Theorem 2.3.10. *Suppose X is an M -valued semimartingale (up to a stopping time τ), and U_0 is an $F(M)$ -valued $\mathcal{F}_0$ -random variable such that $\pi(U_0) = X_0$. Then there exists a unique horizontal lift U (up to τ) of X starting from U_0 .*

Proof. Consider a SDE defined as

$$dU_t = P_\alpha^H(U_t) \circ dX_t^\alpha.$$

Let U_t be its unique solution, so U_t is clearly a horizontal lift since $P_\alpha^H \in \Gamma F(M)$. It suffices to show $\pi(U_t) = X_t$. Let

$$Y_t = \widetilde{f}(U_t) = f(\pi(U_t)) = \pi(U_t) \in M \subset \mathbb{R}^N.$$

By above, we have

$$dY_t = P_\alpha^H \widetilde{f}(U_t) \circ dX_t^\alpha = P_\alpha(Y_t) \circ dX_t^\alpha.$$

By above lemma, we know $dX_t = P_\alpha(X_t) \circ dX_t^\alpha$. By the uniqueness of SDE, $X = Y$.

Next, assume there exists another horizontal lift Π_t of X_t . Then there exists a semimartingale W such that

$$d\Pi_t = H_i(\Pi_t) \circ dW_t^i.$$

and by above theorem

$$dW_t = \Pi_t^{-1} P_\alpha(X_t) \circ dX_t^\alpha.$$

So

$$\begin{aligned} d\Pi_t &= (\Pi_t^{-1} P_\alpha(X_t))^i H_i(\Pi_t) \circ dX_t^\alpha \\ &= P_\alpha^H(\Pi_t) \circ dX_t^\alpha. \end{aligned}$$

So $\Pi = U$ by the uniqueness of solution of SDE. □

Remark 2.3.11. When considering explosion time, there is a fact, if X on M is a semimartingale up to τ , then its horizontal lift H on $F(M)$ is also defined up to τ .

Similarly as the deterministic case, given a semimartingale X on M , the horizontal lift U_t provide the stochastic parallel moving along X_t ,

$$\tau_{t_1 t_2}^X = U_{t_2} U_{t_1}^{-1}: T_{X_{t_1}} M \rightarrow T_{X_{t_2}} M.$$

Proposition 2.3.12. *Let a semimartingale X on a manifold M be the solution of $\text{SDE}(V_1, \dots, V_N; Z, X_0)$ and V_α^H be the horizontal lift of V_α . Then the horizontal lift U of X starting from U_0 satisfy $\text{SDE}(V_1^H, \dots, V_N^H; Z, U_0)$, and the anti-development of X is given by*

$$W_t = \int_0^t U_s^{-1} V_\alpha(X_s) \circ dZ_s^\alpha.$$

Proof. Assume $M \subset \mathbb{R}^N$ that is closed and use above notations. First, we similar have

$$V_\alpha^H \tilde{f}(u) = V_\alpha(\pi(u)).$$

Let Π be the solution of $\text{SDE}(V_1^H, \dots, V_N^H; Z, U_0)$ and $Y_t = \tilde{f}(\Pi_t) = \pi(\Pi_t)$.

$$dY_t = V_\alpha^H \tilde{f}(\Pi_t) \circ dZ_t^\alpha = V_\alpha(Y_t) \circ dZ_t^\alpha,$$

which implies that $Y = X$ by the uniqueness of solution of $\text{SDE}(V_1, \dots, V_N; Z, X_0)$. And by the uniqueness of horizontal lift, $U = \Pi$.

Next, the anti-development of X is given by

$$dW_t = U_t^{-1} P_\beta(X_t) \circ dX_t^\beta.$$

Because $dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$, i.e., $dX_t^\beta = V_\alpha^\beta(X_t) \circ dZ_t^\alpha$,

$$dW_t = U_t^{-1} P_\beta(X_t) V_\alpha^\beta(X_t) \circ dZ_t^\alpha.$$

Because $V_\alpha(X_t) \in T_{X_t} M$,

$$P_\beta(X_t) V_\alpha^\beta(X_t) = P(X_t) V_\alpha(X_t) = V_\alpha(X_t).$$

So

$$dW_t = U_t^{-1} V_\alpha(X_t) \circ dZ_t^\alpha. \quad \square$$

2.4 Stochastic Line Integral

Line Integral. For any 1-form $\omega = f(t)dt$ on $[a, b]$ of $\mathbb{R}$, define

$$\int_{[a,b]} \omega = \int_a^b f(t)dt.$$

Let M be a smooth manifold and $\omega \in \Gamma(T^*M)$. Let $\gamma: [a, b] \rightarrow M$ smooth. Define the line integral of ω along γ as

$$\int_\gamma \omega = \int_{[a,b]} \gamma^* \omega.$$

Clearly, it satisfies the linearity in ω and additivity w.s.t. γ . Moreover, for any smooth $F: M \rightarrow N$ and $\eta \in \Gamma(T^*N)$ and γ in M ,

$$\int_\gamma F^* \eta = \int_{F \circ \gamma} \eta.$$

Proposition 2.4.1. Let $\omega \in \Gamma(T^*M)$ and $\gamma: [a, b] \rightarrow M$ smooth.

$$\int_{\gamma} \omega = \int_a^b \omega_{\gamma(t)}(\dot{\gamma}(t)) dt$$

In particular, if $\omega = df$ for some $f \in C^\infty(M)$,

$$\int_{\gamma} df = \int_a^b (f \circ \gamma)'(t) dt = f(\gamma(b)) - f(\gamma(a)).$$

Proof. Let $\gamma(t) = (\gamma^1(t), \dots, \gamma^d(t))$ and $\omega = \omega_i dx^i$ be the coordinate expression. Then

$$(\gamma^* \omega)_t = \omega_i(\gamma(t)) dx^i(\gamma(t)) = \omega_i(\gamma(t)) d\gamma^i(t) = \omega_i(\gamma(t)) (\gamma^i)'(t) dt$$

On the other hand,

$$\omega_{\gamma(t)}(\dot{\gamma}(t)) = \omega_i(\gamma(t)) dx^i \left((\gamma^j)'(t) \frac{\partial}{\partial x^j} \right) = \omega_i(\gamma(t)) (\gamma^i)'(t).$$

Therefore, we get the desired result. $\square$

Stochastic Line Integral. Let $x: [0, t] \rightarrow M$ and $u(t)$ be its horizontal lift on $F(M)$ starting from u_0 . Let $w(t)$ be the anti-development of $x(t)$, i.e., $\dot{w}_t = u_t^{-1} \dot{x}_t$ and $w_0 = 0$. Let $\{e_i\}$ be the canonical basis of $\mathbb{R}^d$ and so $\{u_t e_i\}$ is a basis of $T_{x_t} M$. Let

$$w_t = w_t^i e_i \Rightarrow \dot{x}_t = \dot{w}_t^i (u_t e_i).$$

Let $\theta \in \Gamma(t^*M)$. Then

$$\int_x \theta = \int_0^t \theta_{x_s}(\dot{x}_s) ds = \int_0^t \dot{w}_s^i \theta_{x_s}(u_s e_i) ds.$$

Definition 2.4.2. Let $\theta \in \Gamma(t^*M)$ and X be an M -valued semimartingale. Let U be a horizontal lift of X and W be its anti-development. The stochastic line integral of θ along X is defined by

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta_{X_s}(U_s e_i) \circ dW_s^i.$$

Remark 2.4.3. For such θ , let $\tilde{\theta}$ be the scalarization of 1-form θ . Then by the definition, $\tilde{\theta}: F(M) \rightarrow \mathbb{R}^d$ by viewing $(\mathbb{R}^d)^* = \mathbb{R}^d$ and

$$\tilde{\theta}(u) = (\theta_{\pi(u)}(u e_1), \dots, \theta_{\pi(u)}(u e_d))^{\top} \in \mathbb{R}^d.$$

So

$$\int_{X_{[0,t]}} \theta = \int_0^t \tilde{\theta}(U_s)_i \circ dW_s^i = \int_0^t \tilde{\theta}(U_s)^{\top} \circ dW_s.$$

Note that $\int_{X_{[0,t]}} \theta$ seems depend on the choice of horizontal lift, and so the connection on M . Actually, it is independent of the above choices by the following proposition and the fact that any semimartingale on M is a solution of some SDE shown in above section.

Proposition 2.4.4. Let $\theta \in \Gamma(T^*M)$ and X be the solution of $dX_t = V_{\alpha}(X_t) \circ dZ_t^{\alpha}$. Then

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta(V_{\alpha})(X_s) \circ dZ_s^{\alpha}.$$

Proof. Choose a horizontal lift U and corresponding anti-development W of X . Then

$$dW_t = U_t^{-1} V_\alpha(X_t) \circ dZ_t^\alpha.$$

Therefore,

$$\begin{aligned} \tilde{\theta}(U_s)_i \circ dW_s^i &= \tilde{\theta}(U_s)_i (U_s^{-1} V_\alpha(X_s))^i \circ dZ_s^\alpha \\ &= \theta_{X_s}(U_s e_i) (U_s^{-1} V_\alpha(X_s))^i \circ dZ_s^\alpha \\ &= \theta_{X_s}(U_s (e_i (U_s^{-1} V_\alpha(X_s))^i)) \circ dZ_s^\alpha \\ &= \theta_{X_s}(V_\alpha(X_s)) \circ dZ_s^\alpha. \end{aligned}$$

Therefore, we have

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta(V_\alpha)(X_s) \circ dZ_s^\alpha. \quad \square$$

Remark 2.4.5. Note that the stochastic line integral of 1-form θ along a given M -valued semimartingale X is independent of the choice of horizontal lift and also the connection. Symbolically,

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta \circ dX_t$$

Example 2.4.6. (1) For all $f \in C^\infty(M)$, assume $dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$.

$$\int_{X_{[0,t]}} df = \int_0^t V_\alpha f(X_s) \circ dZ_s^\alpha = f(X_t) - f(X_0)$$

(2) Let $(U, x^1, \dots, x^d)$ be a local chart of M . Let $\theta = \theta_i dx^i$. For any X a semimartingale on M , let $X^i = x^i(X)$ that is a semimartingale on $\mathbb{R}$.

$$dX_t = V_i(X_t) \circ dX_t^i, \quad V_i = \frac{\partial}{\partial x^i},$$

because $df(X_t) = \frac{\partial f}{\partial x^i}(X_t) \circ dX_t^i$. Then

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta(V_i(X_s)) \circ dX_s^i = \int_0^t \theta_i(X_s) \circ dX_s^i,$$

which is the local expression of stochastic line integral.

Definition 2.4.7. Let $\theta: \Gamma(TF(M)) \rightarrow C^\infty(F(M), \mathbb{R}^d)$, i.e., a $\mathbb{R}^d$ -valued 1-form on $F(M)$. Define

$$\theta(Z)(u) = \theta_u(Z_u) := u^{-1}(d\pi_u(Z_u)),$$

for any $Z \in \Gamma(TF(M))$ and $u \in F(M)$, where $\pi: F(M) \rightarrow M$ is the natural projection. Then θ is called the solder form.

Proposition 2.4.8. Let U be a horizontal semimartingale on $F(M)$. Then its the corresponding anti-development is given by

$$W_t = \int_{U_{[0,t]}} \theta,$$

where θ is the solder form.

Proof. Let $X_t = \pi(U_t)$. Then we have

$$dU_t = P_\alpha^H(U_t) \circ dX_t^\alpha.$$

Note that $\pi_* P_\alpha^H = P_\alpha$. So we get

$$\begin{aligned} \int_{U_{[0,t]}} \theta &= \int_0^t \theta(P_\alpha^H)(U_s) \circ dX_s^\alpha \\ &= \int_0^t U_s^{-1} d\pi_{U_s}(P_\alpha^H(U_s)) \circ dX_s^\alpha \\ &= \int_0^t U_s^{-1} P_\alpha(\pi(U_s)) \circ dX_s^\alpha = W_t. \end{aligned}$$

□

Quadratic Variation. Let $h \in \Gamma(\otimes^{(0,2)} TM)$ and $\tilde{h}$ be its scalarization. So

$$\tilde{h}(u)(e, e') = h_{\pi(u)}(ue, ue'), \quad \forall e, e' \in \mathbb{R}^d, u \in F(M).$$

Let

$$h^{\text{sym}}(e, e') = \frac{h(e, e') + h(e', e)}{2}.$$

Definition 2.4.9. Let $h \in \Gamma(\otimes^{(0,2)} TM)$ and X be an M -valued semimartingale. Let U be a horizontal lift of X and W be its anti-development. Note that

$$dX_t = U_t e_i \circ dW_t^i,$$

Then the h -quadratic variation of X is defined as

$$\begin{aligned} \int_0^t h(dX_s, dX_s) &:= \int_0^t h_{X_s}(U_s e_i, U_s e_j) d\langle W^i, W^j \rangle_s \\ &= \int_0^t \tilde{h}(U_s)(e_i, e_j) d\langle W^i, W^j \rangle_s. \end{aligned}$$

By viewing $dW_s^i dW_s^j = d\langle W^i, W^j \rangle_s$, then

$$\int_0^t h(dX_s, dX_s) = \int_0^t \tilde{h}(U_s)(dW_s, dW_s).$$

Obviously,

$$\int_0^t h(dX_s, dX_s) = \int_0^t h^{\text{sym}}(dX_s, dX_s)$$

and if h is anti-symmetric, $\int_0^t h(dX_s, dX_s) = 0$.

Remark 2.4.10. Note that this definition is independent the choice of U , i.e., the choice of U_0 , because the anti-development of $U_t g$ is $g^{-1}W_t$ and

$$\begin{aligned} h(U_t g e_i, U_t g e_j) d\langle (g^{-1}W)^i, (g^{-1}W)^j \rangle_t &= \tilde{h}(U_t)(g e_i, g e_j) d\langle (g^{-1}W)^i, (g^{-1}W)^j \rangle_t \\ &= g_i^\alpha g_j^\beta \tilde{h}(U_t)(e_\alpha, e_\beta) \tilde{g}_k^i \tilde{g}_\ell^j d\langle W^k, W^\ell \rangle_t \\ &= \tilde{h}(U_t)(e_k, e_\ell) d\langle W^k, W^\ell \rangle_t, \end{aligned}$$

where $g e_i = g_i^j$ and $(\tilde{g}_i^j) = (g_i^j)^{-1}$. So h -quadratic variation is well-defined.

Proposition 2.4.11. Let $h \in \Gamma(\otimes^{(0,2)}TM)$ and X be the solution of $\text{SDE}(V_1, \dots, V_N; Z, X_0)$. Then

$$\int_0^t h(dX_s, dX_s) = \int_0^t h(V_\alpha, V_\beta)(X_s) d\langle Z^\alpha, Z^\beta \rangle_s.$$

Proof. Note that

$$dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$$

Let U be a horizontal lift of X and W be its anti-development. Then

$$dW_t = U_t^{-1} V_\alpha(X_t) \circ dZ_t^\alpha.$$

and so

$$d\langle W^i, W^j \rangle_t = (U_t^{-1} V_\alpha(X_t))^i (U_t^{-1} V_\beta(X_t))^j d\langle Z^\alpha, Z^\beta \rangle_t.$$

It implies that

$$\begin{aligned} h(U_t e_i, U_t e_j) d\langle W^i, W^j \rangle_t &= h(U_t e_i, U_t e_j) (U_t^{-1} V_\alpha(X_t))^i (U_t^{-1} V_\beta(X_t))^j d\langle Z^\alpha, Z^\beta \rangle_t \\ &= h(V_\alpha, V_\beta)(X_t) d\langle Z^\alpha, Z^\beta \rangle_t. \end{aligned} \quad \square$$

Example 2.4.12. (1) Let $(U, x^1, \dots, x^d)$ be a coordinate and $h \in \Gamma(\otimes^{(0,2)}TM)$ locally written as

$$h = h_{ij} dx^i \otimes dx^j.$$

For $X = (X^i)$ on M ,

$$\int_0^t h(dX_s, dX_s) = \int_0^t h_{ij}(X_s) d\langle X^i, X^j \rangle_s,$$

by $dX_t = \frac{\partial}{\partial x^i}(X_t) \circ dX_t^i$.

(2) For $f \in C^\infty(M)$, $\nabla^2 f \in \Gamma(\otimes^{(0,2)}TM)$ and so

$$\begin{aligned} \int_0^t \nabla^2 f(dX_s, dX_s) &= \int_0^t \nabla^2 f_{X_s}(U_s e_i, U_s e_j) d\langle W^i, W^j \rangle_s \\ &= \int_0^t H_j H_i \tilde{f}(U_s) d\langle W^i, W^j \rangle_s. \end{aligned}$$

(3) If $f, g \in C^\infty(M)$,

$$\int_0^t (df \otimes dg)(dX_s, dX_s) = \langle f(X), g(X) \rangle_t.$$

Proof. For X an M -valued semimartingale, WTLG assume $dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$. Then

$$\int_0^t (df \otimes dg)(dX_s, dX_s) = \int_0^t V_\alpha f(X_s) V_\beta f(X_s) d\langle Z^\alpha, Z^\beta \rangle_s.$$

On the other hand, because

$$df(X_t) = V_\alpha f(X_t) \circ dZ_t^\alpha, \quad dg(X_t) = V_\beta g(X_t) \circ dZ_t^\beta,$$

we have

$$\langle f(X), g(X) \rangle_t = \int_0^t V_\alpha f(X_s) V_\beta f(X_s) d\langle Z^\alpha, Z^\beta \rangle_s. \quad \square$$

Proposition 2.4.13. *If $\theta \in \Gamma(\otimes^{(0,2)}TM)$ is positive semi-definite, then the θ -quadratic variation of a semimartingale X is nondecreasing.*

Proof. First,

$$\int_0^t h(dX_s, dX_s) = \int_0^t h(U_s e_i, U_s e_j) d\langle W^i, W^j \rangle_s.$$

Because h is positive semi-positive, the matrix $(h(U_s e_i, U_s e_j))$ is positive semi-definite and let $(m_i^k(s))$ be its squared root matrix. Let

$$J_t^k = \int_0^t m_i^k(s) dW_s^i.$$

Therefore,

$$\int_0^t h(dX_s, dX_s) = \sum_k \int_0^t m_i^k(s) m_j^k(s) d\langle W^i, W^j \rangle_s = \sum_k \langle J^k, J^k \rangle_t. \quad \square$$

2.5 Martingale on Manifold

Definition 2.5.1. Suppose M is a C^∞ manifold with a connection ∇ . An M -valued semimartingale X is called a ∇ -martingale if its anti-development W w.s.t ∇ is an $\mathbb{R}^d$ -valued local martingale.

Remark 2.5.2. It is well defined because if W and W' are two anti-development of X , then there exists a $g \in GL(d, \mathbb{R})$ such that $W' = gW$.

Remark 2.5.3. Note that when $M = \mathbb{R}^N$ is equipped with the canonical connection. Then for any $c(t)$ on M , its horizontal lift starting from u_0 is

$$u(t)a = (u_0)_j^i a^j \frac{\partial}{\partial x^i} \Big|_{c(t)}.$$

When viewing $T_x M \subset \mathbb{R}^N$, $u(t)a \equiv u_0 a$. Define $f: F(M) \rightarrow \mathbb{R}^N$ as $f(u) = u e_i$. Then

$$P_\alpha^H(u)f = \frac{d}{dt} \Big|_{t=0} f(u_t) = \frac{d}{dt} \Big|_{t=0} u_t e_i = 0.$$

Let X be an M -valued semimartingale. Because $dU_t = P_\alpha^H(U_t) \circ dX_t^\alpha$,

$$df(U_t) = P_\alpha^H f(U_t) \circ dX_t^\alpha = 0 \Rightarrow f(U_t) = U_t e_i = U_0 e_i.$$

Therefore,

$$\begin{aligned} X_t &= X_0 + \int_0^t U_s e_i \circ dW_s^i \\ &= X_0 + \int_0^t U_0 e_i dW_s^i \\ &= X_0 + U_0 W_t. \end{aligned}$$

It follows that when $M = \mathbb{R}^N$, if X is an M -martingale, then it is a local martingale in the usual sense.

Proposition 2.5.4. *An M -valued semimartingale X is a ∇ -martingale if and only if*

$$N^f(X)_t := f(X_t) - f(X_0) - \frac{1}{2} \int_0^t \nabla^2 f(dX_s, dX_s),$$

is an $\mathbb{R}$ -valued local martingale for every $f \in C^\infty(M)$.

Remark 2.5.5. For Euclidean case, a continuous semimartingale X is a local martingale if and only if $N^f(X)$ is a local martingale for all $f \in C^\infty$ (or $f \in C_c^\infty$) by Itô formula.

Proof. $\Rightarrow$: Let U be a horizontal lift of X and W be its anti-development. Then

$$dU_t = H_i(U_t) \circ dW_t^i.$$

For $f \in C^\infty(M)$, let $\tilde{f} = f \circ \pi \in C^\infty(F(M))$. So

$$\begin{aligned} f(X_t) - f(X_0) &= \tilde{f}(U_t) - \tilde{f}(U_0) \\ &= \int_0^t H_i \tilde{f}(U_s) \circ dW_s^i \\ &= \int_0^t H_i \tilde{f}(U_s) dW_s^i + \frac{1}{2} \left\langle H_i \tilde{f}(U), W^i \right\rangle_t. \end{aligned}$$

Note that

$$dH_i \tilde{f}(U_t) = H_j H_i \tilde{f}(U_t) \circ dW_t^j,$$

which implies that

$$\begin{aligned} \left\langle H_i \tilde{f}(U), W^i \right\rangle_t &= \int_0^t H_j H_i \tilde{f}(U_s) d\langle W^i, W^j \rangle_s \\ &= \int_0^t \nabla^2 f(dX_s, dX_s). \end{aligned}$$

Therefore,

$$N^f(X)_t = \int_0^t H_i \tilde{f}(U_t) dW_t^i.$$

When X is an M -local semimartingale, i.e., W is an $\mathbb{R}^d$ -local martingale, then $N^f(X)$ is obviously a local martingale.

$\Leftarrow$: Assume $M \subset \mathbb{R}^N$ properly embedded. Let $f: M \rightarrow \mathbb{R}^N$ be the coordinate function $f(x) = x$ and $\tilde{f} = f \circ \pi$ be its lift on $F(M)$. When viewing $T_x M \subset \mathbb{R}^N$, we already have $H_i \tilde{f}(u) = u e_i$. So similarly as above calculation,

$$\begin{aligned} N^f(X)_t &:= N^{f^\alpha}(X)_t e_\alpha \\ &= e_\alpha \int_0^t H_i \tilde{f}^\alpha(U_s) dW_s^i \\ &= e_\alpha \int_0^t (U_s e_i)^\alpha dW_s^i \\ &= \int_0^t U_s e_i dW_s^i = \int_0^t U_s dW_s, \end{aligned}$$

by viewing $U_s: \mathbb{R}^d \rightarrow \mathbb{R}^N$ where $U_s: \mathbb{R}^d \rightarrow T_x M$ is an isomorphism. Define $V_s: \mathbb{R}^N = T_x M \oplus N_x M \rightarrow \mathbb{R}^d$ as

$$V_s \xi = \begin{cases} U_s^{-1} \xi, & \xi \in T_{X_s} M, \\ 0, & \xi \perp T_{X_s} M \end{cases}$$

Then $V_s U_s e = e$ for all $\mathbb{R}^d$. Therefore,

$$\int_0^t V_s dN^f(X)_s = \int_0^t V_s U_s dW_s = W_t.$$

Because $N^f(X)$ is a local martingale, W_t is a local martingale. It follows that X is an M -martingale. $\square$

Remark 2.5.6. From the proof, we can see that when viewing $M \subset \mathbb{R}^N$ and let $f = (f^1, \dots, f^N): M \rightarrow \mathbb{R}^N$ be the coordinate function, if N^{f^α} for $\alpha = 1, \dots, N$ are local martingales, then X is an M -local martingale.

Proposition 2.5.7. *Suppose $(U, x^1, \dots, x^d)$ is a local chart on M and $X = (X^i)$, i.e. $X^i = x^i(X)$, a semimartingale on M . Then X is an M -martingale if and only if*

$$N_t^i = X_t^i - X_0^i + \frac{1}{2} \int_0^t \Gamma_{jk}^i(X_s) d\langle X^j, X^k \rangle_s$$

is a local martingale.

Proof. $\Rightarrow$: Locally, we know

$$\nabla^2 x^i = -\Gamma_{jk}^i dx^j \otimes dx^k,$$

which implies that

$$\int_0^t \nabla^2 x^i(dX_s, dX_s) = - \int_0^t \Gamma_{jk}^i(X_s) d\langle X^j, X^k \rangle_s.$$

Therefore, this direction can be directly obtained by applying above theorem to each $f = x^i$.
 $\Leftarrow$: Choose a local coordinate for $F(M)$ as $(\tilde{x}^i, e_j^i)$, where

$$\tilde{x}^i(u) = x^i(\pi(u)), \quad e_j^i(u) \text{ for } u e_j = e_j^i(u) \frac{\partial}{\partial x^j} \Big|_{\pi(u)}.$$

Let U be a horizontal lift with the anti-development W of X . Then

$$dU_t = H_j(U_t) \circ dW_t^j.$$

Because $X_t = \pi(U_t)$ and $U_t e_j = e_j^i(t) \frac{\partial}{\partial x^j} \Big|_{X_t}$, where $e_j^i(t) = e_j^i(U_t)$, we have

$$\begin{aligned} dX_t^i &= d\tilde{x}^i(U_t) = H_j \tilde{x}^i(U_t) \circ dW_t^j \\ &= (U_t e_j)(x^i) \circ dW_t^j \\ &= e_j^i(t) \circ dW_t^j. \end{aligned}$$

Moreover, we also know

$$dU_t = P_k^H(U_t) \circ dX_t^k.$$

Note that in the local chart, M can be viewed as $\mathbb{R}^d$ and so $P(x): \mathbb{R}^d \rightarrow T_x M$ the orthogonal projection is the identity, which implies that

$$P_k(X_t) = P(X_t) e_k = e_k = \delta_k^\ell \frac{\partial}{\partial x^\ell} \Big|_{X_t}.$$

Let $v(s)$ be the horizontal lift starting from U_t of a curve $c(s)$ with $c(0) = U_t$ and $\dot{c}(0) = P_k(X_t) = \delta_k^m \frac{\partial}{\partial x^m} \Big|_{X_t}$. Then by using the coordinates, we have

$$\dot{v}(0) = P_k^H(U_t) = -\Gamma_{m\ell}^i(X_t) \delta_k^m e_j^\ell(t) \frac{\partial}{\partial e_j^i} \Big|_{U_t} + \delta_k^m \frac{\partial}{\partial \tilde{x}^m} \Big|_{U_t}.$$

So

$$P_k^H e_j^i(U_t) = -\Gamma_{k\ell}^i(X_t) e_j^\ell(t).$$

It follows that

$$de_j^i(t) = -\Gamma_{k\ell}^i(X_t) e_j^\ell(t) \circ dX_t^k.$$

Let $(f_j^i) = (e_j^i)^{-1}$. Then we get

$$dW_t^j = f_k^j(t) \circ dX_t^k.$$

So

$$\begin{aligned} dX_t^i &= e_j^i(t) dW_t^j + \frac{1}{2} d\langle e_j^i, W^j \rangle_t \\ &= e_j^i(t) dW_t^j - \frac{1}{2} \Gamma_{k\ell}^i(X_t) e_j^\ell(t) f_m^j(t) d\langle X^k, X^m \rangle_t \\ &= e_j^i(t) dW_t^j - \frac{1}{2} \Gamma_{k\ell}^i(X_t) d\langle X^k, X^\ell \rangle, \end{aligned}$$

which implies that

$$dN_t^i = e_j^i(t) dW_t^j \Rightarrow dW_t^i = f_j^i(t) dN_t^j.$$

Therefore, if N_t^i are all local martingales, so is W_t and X is an M -valued martingale. $\square$

2.6 Martingales on Submanifolds

Let M, N be two manifolds equipped with connection ∇^M, ∇^N respectively, and $M \subset N$ be a submanifold. Consider the second fundamental form $\text{II}: \Gamma(TM) \times \Gamma(TM) \rightarrow \Gamma(TN|_M)$ defined as

$$\text{II}(V, W) = \nabla_V^N W - \nabla_V^M W,$$

which is obviously bilinear over $C^\infty(M)$, so II is a $TN|_M$ -valued $(0, 2)$ -tensor field on M .

Remark 2.6.1. A direct result of this is, $\text{II}(V, W)(p)$ is fully determined by V_p and W_p

Moreover, for a connection ∇ , let

$$T(V, W) = \nabla_V W - \nabla_W V - [V, W].$$

We have

$$\text{II}(V, W) - \text{II}(W, V) = T^N(V, W) - T^M(V, W),$$

so II is symmetric if ∇^M, ∇^N are torsion-free. If $\text{II} \equiv 0$, then M is called totally geodesic.

Proposition 2.6.2. *If M is totally geodesic, then any ∇^M -geodesic in M is also a ∇^N -geodesic in N . When ∇^M, ∇^N are torsion-free, the converse is also true.*

Lemma 2.6.3. *Let $M \subset \mathbb{R}^N$ be a submanifold and $f: M \rightarrow \mathbb{R}^N$ be the coordinate function on M with lift $\tilde{f} = f \circ \pi$ to the frame bundle $F(M)$. Then*

$$H_i H_j \tilde{f}(u) = \text{II}(ue_i, ue_j)$$

Proof. Let u_t be the horizontal lift starting from $u_0 = u$ of a curve x_t with $x_0 = \pi(u)$ and $\dot{x}_0 = ue_i$, i.e., $\dot{u}(0) = H_i(u)$, and $V(x_t) = u_t e_j$ is a ∇^M -parallel along x_t . So

$$\nabla_{\dot{x}_t}^M V = 0 \Rightarrow \nabla_{ue_i}^M V = 0.$$

It implies that

$$\Pi(ue_i, ue_j) = \nabla_{ue_i}^{\mathbb{R}^N} V.$$

On the other hand, recall when viewing $T_x M \subset \mathbb{R}^N$,

$$H_j \tilde{f}(u_t) = u_t e_j = V(x_t).$$

Therefore,

$$H_i H_j \tilde{f}(u) = \left. \frac{d}{dt} \right|_{t=0} V(x_t) = \nabla_{ue_i}^{\mathbb{R}^N} V = \Pi(ue_i, ue_j)$$

for viewing $V(x_t) = (V^1(x_t), \dots, V^N(x_t)) \in \mathbb{R}^N$. $\square$

Theorem 2.6.4. *Using notations as above and assume $\dim M = l$ and $\dim N = d$. Suppose that X is a semimartingale on $M \subset N$. Let U^M be a horizontal lift of X in $F(M)$ with the corresponding $\mathbb{R}^d$ -valued anti-development W^M , and U^N be a horizontal lift of X in $F(N)$ with the corresponding $\mathbb{R}^l$ -valued anti-development W^N . Then*

$$W_t^N = \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \frac{1}{2} \int_0^t (U_s^N)^{-1} \Pi(dX_s, dX_s).$$

Proof. First,

$$\Pi(dX_s, dX_s) = \Pi(U_s^M e_i, U_s^M e_j) d\langle (W^M)^i, (W^M)^j \rangle_s.$$

So we need to show

$$W_t^N = \int_0^t (U_s^N)^{-1} U_s^M e_i d(W^M)_s + \int_0^t (U_s^N)^{-1} \Pi(U_s^M e_i, U_s^M e_j) d\langle (W^M)^i, (W^M)^j \rangle_s.$$

First, assume $N = \mathbb{R}^l$. WTLG, we can choose $U_0^N = I$. In such case, we have

$$W^N = X - X_0, \quad U^N \equiv I.$$

For simplicity, denote $U = U^M$ and $W = W^M$. Let $f: M \rightarrow \mathbb{R}^l$ be the coordinate map and $\tilde{f} = f \circ \pi$. Then we have been shown

$$X_t = X_0 + \int_0^t H_i \tilde{f}(U_s) dW_s^i + \frac{1}{2} \int_0^t H_i H_j \tilde{f}(U_s) d\langle W^i, W^j \rangle_s.$$

Note that

$$H_i \tilde{f}(u) = ue_i, \quad H_i H_j \tilde{f}(u) = \Pi(ue_i, ue_j).$$

Therefore,

$$W_t^N = X_t - X_0 = \int_0^t U_s dW_s + \frac{1}{2} \int_0^t \Pi(dX_s, dX_s).$$

Next, for general N , assume $N \subset \mathbb{R}^N$ embedded manifold. Let Π^M and Π^N be the second fundamental forms of M and N in $\mathbb{R}^N$ respectively. Then

$$X_t = X_0 + \int_0^t U_s^M dW_s^M + \frac{1}{2} \int_0^t \Pi^M(dX_s, dX_s)$$

$$X_t = X_0 + \int_0^t U_s^N dW_s^N + \frac{1}{2} \int_0^t \Pi^N(dX_s, dX_s),$$

where the two equations are viewed in $\mathbb{R}^N$, $T_x M = \mathbb{R}^l$ and $T_x N = \mathbb{R}^d$. Note that

$$\Pi(X, Y) = \Pi^M(X, Y) - \Pi^N(X, Y).$$

By above, $dX_t = U_s^N dW_s^N + \frac{1}{2} \Pi^N(dX_s, dX_s)$,

$$\begin{aligned} W_t^N &= \int_0^t (U_s^N)^{-1} dX_t - \frac{1}{2} \int_0^t (U_s^N)^{-1} \Pi^N(dX_s, dX_s) \\ &= \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \frac{1}{2} \int_0^t (U_s^N)^{-1} \Pi^M(dX_s, dX_s) \\ &\quad - \frac{1}{2} \int_0^t (U_s^N)^{-1} \Pi^N(dX_s, dX_s) \\ &= \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \frac{1}{2} \int_0^t (U_s^N)^{-1} \Pi(dX_s, dX_s). \end{aligned} \quad \square$$

Corollary 2.6.5. *Suppose that $M \subset N$ is a totally geodesic submanifold of N . Then every ∇^M -martingale on M is also a ∇^N -martingale on N .*

For $x \in M$, define

$$\text{ran } \Pi_x := \{\Pi(X, Y)(x) : X, Y \in \Gamma(TM)\} \subset T_x N.$$

Π is called independent of TM if $\text{ran } \Pi_x \cap T_x M = \{0\}$ for every $x \in M$. Note that for Riemannian submanifold, Π is obviously independent of TM .

Corollary 2.6.6. *Suppose that $M \subset N$ such that Π is independent of TM . Then a ∇^N -martingale on N which lives on M is also a ∇^M -martingale on M .*

Proof. First, W^M is clearly a continuous semi-martingale, so it can be uniquely decomposed into

$$W_t = M_t + A_t,$$

where M is a local martingale and A is a process of bounded variation with $A_0 = 0$. It suffices to prove $A = 0$. By above,

$$W_t^N = \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \int_0^t (U_s^N)^{-1} \left(U_s^M dA_s + \frac{1}{2} \Pi(dX_s, dX_s) \right).$$

Because W^N is a local martingale,

$$\int_0^t (U_s^N)^{-1} \left(U_s^M dA_s + \frac{1}{2} \Pi(dX_s, dX_s) \right) = 0$$

and so

$$U_s^M dA_s + \frac{1}{2} \Pi(dX_s, dX_s) = 0.$$

However,

$$U_s^M dA_s \in T_{\pi(U_s^M)} M, \quad \Pi(dX_s, dX_s) \in \text{ran } \Pi_{\pi(U_s^M)}.$$

The independence implies that $dA_s = 0$, and so $A = 0$. $\square$

Example 2.6.7. (1) Let $M \subset N$ be a Riemannian hypersurface. It is called (strictly) convex if the scalar fundamental form is (strictly) positive definite at each point. Assume M is strictly convex. Because Π is independent of TM , any ∇^N -martingale on N which lies on M is also a ∇^M -martingale. It implies that

$$\Pi(dX_s, dX_s) = 0.$$

However, because of the strict positivity, $dX_s = 0$, and so X is constant.

(2) Let M be a Riemannian manifold of dimension $\mathbb{R}^d$. A semimartingale X on M is called a Brownian motion if its anti-development W is a d -dimensional standard Brownian motion. Let $M \subset \mathbb{R}^N$ be a minimal submanifold, i.e., the mean curvature field $H = \text{tr}_g \Pi = 0$. Then a Brownian motion on M is a local martingale on $\mathbb{R}^N$, because

$$\Pi(U_t e_i, U_t e_j) d\langle W^i, W^j \rangle_t = \sum_i \Pi(U_t e_i, U_t e_i) dt = H(X_t) dt = 0.$$

Note that for Riemannian manifold, we consider $O(M)$, and so $\{U_t e_i\}$ is orthonormal in $T_{\pi(U_t)} M$.

Proposition 2.6.8. *Suppose that $M \subset N$ be totally geodesic. For any $x \in M$, there exists a neighborhood O of p in N such that if X is an ∇^N -martingale in O on $[0, T]$ with $X_T \in M$, then $X_t \in M$ for all $t \in [0, T]$.*

Theorem 2.6.9. *Let M be a manifold with a connection. For any $x \in M$, there exists a neighborhood O of x such that: if $(X_t)_{t \in [0, T]}$ and $(Y_t)_{t \in [0, T]}$ are two martingales on V such that $X_T = Y_T$, then $X_t = Y_t$ for all $t \in [0, T]$.*

Chapter 3

Brownian Motion on Manifolds

3.1 Laplace-Beltrami operator

Let (M, g) be a Riemannian manifold equipped with the Levi-Civita connection ∇ and the standard Riemannian volume measure m . For $f, g \in C_c(M)$,

$$(f, g) := \int_M f(x)g(x)m(dx)$$

and for $X, Y \in \Gamma_c(TM)$, i.e., vector fields with compact support,

$$(X, Y) = \int_M \langle X, Y \rangle_x m(dx),$$

and for $\theta, \psi \in \Gamma_c(T^*M)$, i.e., 1-forms with compact support,

$$(\theta, \psi) = \int_M \langle \theta, \psi \rangle_x m(dx).$$

Let Δ_M be the Laplace-Beltrami operator. Because locally

$$\Delta_M f = \text{tr}_g(\nabla^2 f) = g^{ij} \frac{\partial^2}{\partial x^i \partial x^j} f - g^{k\ell} \Gamma_{k\ell}^i \frac{\partial}{\partial x^i} f,$$

Δ_M is a nondegenerate second order elliptic operator. Moreover, by the local expression, for $L = \frac{1}{2}\Delta_M$, the corresponding

$$\Gamma(f, g) = g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial x^j} = \langle \nabla f, \nabla g \rangle.$$

Let $\pi: O(M) \rightarrow M$ be the orthogonal frame bundle. Define a second order differential operator on $O(M)$ as

$$\Delta_{O(M)} := \sum_{i=1}^d H_i^2,$$

called Bochner's horizontal Laplacian.

Proposition 3.1.1. *For any $f \in C^\infty(M)$, let $\tilde{f} = f \circ \pi \in C^\infty(O(M))$. Then for any $u \in O(M)$,*

$$\Delta_M f(x) = \Delta_{O(M)} \tilde{f}(u),$$

where $x = \pi(u)$.

Proof. Fix a $f \in C^\infty(M)$. Note that for any 1-form, when viewing $(\mathbb{R}^d)^* = \mathbb{R}^d$, the scalarization $\tilde{\theta}: O(M) \rightarrow \mathbb{R}^d$ satisfies

$$\tilde{\theta}(u) = (\theta(ue_1), \dots, \theta(ue_d))^\top \in \mathbb{R}^d.$$

Therefore,

$$(\tilde{df})_i = df(ue_i) = ue_i(f) = H_i \tilde{f}(u),$$

and

$$\tilde{df} = \left(H_1 \tilde{f}, H_2 \tilde{f}, \dots, H_d \tilde{f} \right)^\top.$$

Because (ue_i) is a normal basis in $T_{\pi(u)}M$, we have

$$\sharp(ue_i)^* = ue_i.$$

It follows that, first

$$\widetilde{\text{grad } f} = \left(H_1 \tilde{f}, H_2 \tilde{f}, \dots, H_d \tilde{f} \right)^\top.$$

because $\text{grad } f((ue_i)^*) = \langle \text{grad } f, \sharp(ue_i)^* \rangle = \langle \text{grad } f, ue_i \rangle = df(ue_i)$. Second, for any $X \in \Gamma(TM)$,

$$\text{div } X(\pi(u)) = \sum_i \nabla X((ue_i)^*, ue_i) = \sum_i \langle \nabla_{ue_i} X, \sharp(ue_i)^* \rangle = \sum_i \langle \nabla_{ue_i} X, ue_i \rangle.$$

Moreover, since $u \in O(M)$, orthogonal map,

$$\begin{aligned} \text{div } X(\pi(u)) &= \sum_i \langle u^{-1} \nabla_{ue_i} X(\pi(u)), e_i \rangle \\ &= \sum_i \langle \widetilde{\nabla_{ue_i} X}(u), e_i \rangle \\ &= \sum_i \langle H_i \tilde{X}(u), e_i \rangle = \sum_i \left(H_i \tilde{X} \right)^i(u). \end{aligned}$$

It follows that

$$\Delta_M f(\pi(u)) = \text{div}(\text{grad } f)(\pi(u)) = \sum_i \left(H_i(\widetilde{\text{grad } f}) \right)^i(u) = \sum_i H_i(H_i \tilde{f}) = \Delta_{O(M)} \tilde{f}(u). \quad \square$$

Theorem 3.1.2 (Nash's Embedding Theorem). *Every Riemannian manifold can be isometrically embedded in some Euclidean space with the standard metric.*

Remark 3.1.3. Note that if the Riemannian manifold is complete, then we can find an Euclidean space such that the image is closed, i.e., a complete Riemannian manifold can be viewed as a Riemannian submanifold of an Euclidean space, which is also a closed subset.

In the following, we assume a $M \subset \mathbb{R}^N$ be a Riemannian submanifold that is closed. So all above theories can be applied to such M . Similarly as above, let $P(x): \mathbb{R}^N \rightarrow T_x M$ be the orthogonal projection and $P_\alpha = P e_\alpha \in \Gamma(TM)$. Note that if $X \in T_x M$, we have

$$\begin{aligned} X &= P(x)X = P(x) \sum_{\alpha=1}^N \langle X, e_\alpha \rangle_{\mathbb{R}^N} e_\alpha = \sum_{\alpha=1}^N \langle X, e_\alpha \rangle_{\mathbb{R}^N} P_\alpha(x) \\ &= \sum_{\alpha=1}^N \langle P(x)X, e_\alpha \rangle_{\mathbb{R}^N} P_\alpha(x) = \sum_{\alpha=1}^N \langle X, P(x)e_\alpha \rangle_{\mathbb{R}^N} P_\alpha(x) \end{aligned}$$

$$= \sum_{\alpha=1}^N \langle X, P_\alpha(x) \rangle_{T_x M} P_\alpha(x).$$

Let $\tilde{\nabla}$ be the standard Euclidean connection. The induced Levi-Civita connection on M is

$$\nabla_X Y = (\tilde{\nabla}_X Y)^\top = P(\tilde{\nabla}_X Y), \quad X, Y \in \Gamma(TM).$$

Moreover, for any $X \in \Gamma(TM)$ and any $f \in C^\infty(M)$,

$$X(f) = \langle \text{grad } f, X \rangle_{TM} = \langle \overline{\text{grad } f}, X \rangle_{\mathbb{R}^N}, \Rightarrow \text{grad } f = P \overline{\text{grad } f},$$

where $\overline{\text{grad } f}$ is the usual gradient on $\mathbb{R}^N$.

Theorem 3.1.4. *Using notations as above,*

$$\Delta_M = \sum_{\alpha=1}^N P_\alpha^2.$$

Proof. Let $f \in C^\infty(M)$.

$$\text{grad } f = \sum_{\alpha=1}^N \langle \text{grad } f, P_\alpha \rangle P_\alpha = \sum_{\alpha=1}^N P_\alpha(f) P_\alpha.$$

Note that for any $X \in \Gamma(TM)$ and $h \in C^\infty(M)$, it is not difficult such that

$$\text{div}(hX) = X(h) + h \text{div}(X).$$

It follows that

$$\Delta_M f = \sum_{\alpha=1}^N P_\alpha^2(f) + \sum_{\alpha=1}^N P_\alpha(f) \text{div}(P_\alpha).$$

For the second term, fix $x \in M$. First,

$$\text{div}(P_\alpha)(x) = \sum_{i=1}^d \left\langle \nabla_{\frac{\partial}{\partial x^i}} P_\alpha(x), \frac{\partial}{\partial x^i} \Big|_x \right\rangle,$$

where $(x^1, \dots, x^d)$ is a normal coordinate at x , because $\left\{ \frac{\partial}{\partial x^i} \Big|_x \right\}$ is an orthonormal basis of $T_x M$. Moreover, because $\Gamma_{ij}^k(x) = 0$,

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) = 0.$$

So

$$\begin{aligned} \left\langle \nabla_{\frac{\partial}{\partial x^i}} P_\alpha(x), \frac{\partial}{\partial x^i} \Big|_x \right\rangle_{T_x M} &= \frac{\partial}{\partial x^i} \left\langle P_\alpha, \frac{\partial}{\partial x^i} \right\rangle_{TM} (x) = \frac{\partial}{\partial x^i} \left\langle P e_\alpha, \frac{\partial}{\partial x^i} \right\rangle_{TM} (x) \\ &= \frac{\partial}{\partial x^i} \left\langle e_\alpha, P \frac{\partial}{\partial x^i} \right\rangle_{\mathbb{R}^N} (x) = \frac{\partial}{\partial x^i} \left\langle e_\alpha, \frac{\partial}{\partial x^i} \right\rangle_{\mathbb{R}^N} (x) \\ &= \left\langle \tilde{\nabla}_{\frac{\partial}{\partial x^i}} e_\alpha(x), \frac{\partial}{\partial x^i} \right\rangle_{\mathbb{R}^N} + \left\langle e_\alpha(x), \tilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} \\ &= \left\langle e_\alpha, \tilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N}. \end{aligned}$$

It follows that

$$\begin{aligned}
\sum_{\alpha=1}^N \operatorname{div}(P_\alpha)(x) P_\alpha(x) &= \sum_{\alpha=1}^N \sum_{i=1}^d \left\langle e_\alpha, \tilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} P_\alpha(x) \\
&= \sum_{\alpha=1}^N \sum_{i=1}^d \left\langle e_\alpha, \tilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} P(x) e_\alpha \\
&= \sum_{i=1}^d P(x) \sum_{\alpha=1}^N \left\langle e_\alpha, \tilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} e_\alpha \\
&= \sum_{i=1}^d P(x) \tilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) = \sum_{i=1}^d \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) = 0.
\end{aligned}$$

Therefore, $\sum_{\alpha=1}^N P_\alpha(f) \operatorname{div}(P_\alpha) = 0$ for any $f \in C^\infty(M)$. Then

$$\Delta_M f = \sum_{\alpha} P_\alpha^2(f).$$

□

Proposition 3.1.5. *Using same notations as above,*

$$\Delta_M f = \sum_{\alpha} \nabla^2 f(P_\alpha, P_\alpha).$$

Proof. Fix $x \in M$. Let $\{X_i\}$ be an orthonormal basis of $T_x M$ and consider decomposition

$$X_i = \sum_{\alpha=1}^N \langle X_i, P_\alpha(x) \rangle P_\alpha(x).$$

Then

$$\begin{aligned}
\Delta_M f(x) &= \sum_{i=1}^d \nabla^2 f(x)(X_i, X_i) \\
&= \sum_{i=1}^d \sum_{\alpha, \beta=1}^N \nabla^2 f(P_\alpha, P_\beta)(x) \langle X_i, P_\alpha(x) \rangle \langle X_i, P_\beta(x) \rangle \\
&= \sum_{\alpha=1}^N \nabla^2 f \left(P_\alpha, \sum_{i=1}^d \sum_{\beta=1}^N \langle X_i, P_\alpha \rangle \langle X_i, P_\beta \rangle P_\beta \right) (x).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\sum_{i=1}^d \sum_{\beta=1}^N \langle X_i, P_\alpha \rangle \langle X_i, P_\beta \rangle P_\beta &= \sum_{\beta=1}^N \left\langle \sum_{i=1}^d \langle X_i, P_\beta \rangle X_i, P_\alpha \right\rangle P_\beta \\
&= \sum_{\beta=1}^N \langle P_\beta, P_\alpha \rangle P_\beta = P P_\alpha = P_\alpha.
\end{aligned}$$

□

3.2 Brownian Motion on Manifolds

Theorem 3.2.1. *Let $X: \Omega \rightarrow W(M)$ be a measurable map defined on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mu = (X_0)_\# \mathbb{P}$ be the initial distribution on M . TFAE.*

(1) X is a $\frac{1}{2}\Delta_M$ -diffusion process w.s.t. $\mathbb{F}^X$, i.e., X is a $\mathbb{F}^X$ -semimartingale, and

$$M^f(X)_t = f(X_t) - f(X_0) - \frac{1}{2} \int_0^t \Delta_M f(X_s) ds, \quad 0 \leq t < e(X),$$

is an $\mathbb{F}^X$ -local martingale for all $f \in C^\infty(M)$.

(2) X is a $\mathbb{F}^X$ -semimartingale on M whose anti-development is a standard Euclidean $\mathbb{F}^X$ -Brownian motion.

Remark 3.2.2. An M -valued process X satisfies above condition is called a Riemannian Brownian motion on M . Moreover, $\mathbb{P}_\mu = X_\# \mathbb{P}$ is called a Wiener measure on $W(M)$. Note that a Wiener measure is uniquely determined by $\Delta_M/2$ and an initial measure μ .

Proof. $\Leftarrow$: Let U be a horizontal lift of X and W is its anti-development. Then we have

$$dU_t = H_i(U_t) \circ dW_t.$$

For any $f \in C^\infty(M)$, let $\tilde{f} = f \circ \pi$ be its lift on $O(M)$. Then by above mention,

$$f(X_t) = f(X_0) + \int_0^t H_i \tilde{f}(U_s) dW_s^i + \frac{1}{2} \int_0^t H_i H_j \tilde{f}(U_s) d\langle W^i, W^j \rangle_s.$$

Because W is a Brownian motion, $\langle W^i, W^j \rangle_t = \delta^{ij}t$. Then

$$\begin{aligned} \int_0^t H_i H_j \tilde{f}(U_s) d\langle W^i, W^j \rangle_s &= \int_0^t \sum_i H_i^2 \tilde{f}(U_s) ds \\ &= \int_0^t \Delta_{O(M)} \tilde{f}(U_s) ds \\ &= \int_0^t \Delta_M f(X_s) ds. \end{aligned}$$

Therefore,

$$M^f(X)_t = \int_0^t H_i \tilde{f}(U_s) dW_s^i,$$

which is clearly a $\mathbb{F}^X$ -local martingale.

$\Rightarrow$: Let $f = (f^\alpha): M \rightarrow \mathbb{R}^N$ be the coordinate function $f(x) = x$. Let $\widehat{f}^\alpha = f^\alpha \circ \pi$ be the lift on $O(M)$. By assumption, for $X_t^\alpha = f^\alpha(X)_t$,

$$X_t^\alpha = X_0^\alpha + M_t^\alpha + \frac{1}{2} \int_0^t \Delta_M f^\alpha(X_s) ds,$$

where M_t^α is a local martingale, $\alpha = 1, 2, \dots, N$. On the other hand, let H be the horizontal lift of X and W be its anti-development. Then

$$X_t^\alpha = X_0^\alpha + \int_0^t H_i \widehat{f}^\alpha(U_s) dW_s^i + \frac{1}{2} \int_0^t \nabla^2 f^\alpha(dX_s, dX_s).$$

Claim: For any $f \in C^\infty(M)$, we can have

$$\int_0^t \nabla^2 f(dX_s, dX_s) = \int_0^t \Delta_M f(X_s) ds.$$

First, we know $dX_t = P_\alpha(X_t) \circ dX_t^\alpha$. Then we get

$$\int_0^t \nabla^2 f(dX_s, dX_s) = \int_0^t \nabla^2 f(P_\alpha, P_\beta)(X_s) d\langle X^\alpha, X^\beta \rangle_s.$$

By the assumption, because X is a $\Delta_M/2$ -diffusion process,

$$d\langle X^\alpha, X^\beta \rangle_s = d\langle M^\alpha, M^\beta \rangle_s = \Gamma(f^\alpha, f^\beta)(X_s)ds,$$

where

$$\Gamma(f^\alpha, f^\beta) = \langle \nabla f^\alpha, \nabla f^\beta \rangle.$$

Because

$$\text{grad } f^\alpha = P(\overline{\text{grad } f^\alpha}) = P e_\alpha = P_\alpha,$$

we have

$$\Gamma(f^\alpha, f^\beta) = \langle \text{grad } f^\alpha, \text{grad } f^\beta \rangle = \langle P_\alpha, P_\beta \rangle.$$

Moreover,

$$\sum_{\alpha, \beta} \nabla^2 f(P_\alpha, P_\beta) \langle P_\alpha, P_\beta \rangle = \sum_{\alpha} \nabla^2 f(P_\alpha, \sum_{\beta} \langle P_\alpha, P_\beta \rangle P_\beta) = \sum_{\alpha} \nabla^2 f(P_\alpha, P_\alpha) = \Delta_M f$$

Then

$$\begin{aligned} \int_0^t \nabla^2 f(dX_s, dX_s) &= \int_0^t \nabla^2 f(P_\alpha, P_\beta)(X_s) \Gamma(f^\alpha, f^\beta)(X_s) ds \\ &= \int_0^t \sum_{\alpha, \beta} \nabla^2 f(P_\alpha, P_\beta)(X_s) \langle P_\alpha, P_\beta \rangle ds \\ &= \int_0^t \Delta_M f(X_s) ds. \end{aligned}$$

So the claim has been proved.

Then we have

$$M_t^\alpha = \int_0^t H_i \widetilde{f}^\alpha(U_s) dW_s^i.$$

Next, we will use the Lévy's theorem to check W is a Brownian motion. First, because $\widetilde{f}^\alpha = f^\alpha \circ \pi$,

$$\begin{aligned} H_i \widetilde{f}^\alpha(u) &= (ue_i)(f^\alpha) \\ &= \langle ue_i, \text{grad } f^\alpha \rangle_{TM} \\ &= \langle ue_i, Pe_\alpha \rangle_{TM} = \langle ue_i, Pe_\alpha \rangle_{\mathbb{R}^N} \\ &= \langle Pue_i, e_\alpha \rangle_{\mathbb{R}^N} = \langle ue_i, e_\alpha \rangle_{\mathbb{R}^N} \end{aligned}$$

Moreover,

$$\begin{aligned} \sum_{\alpha=1}^N \langle ue_i, e_\alpha \rangle \langle ue_j, e_\alpha \rangle &= \left\langle ue_i, \sum_{\alpha=1}^N \langle ue_j, e_\alpha \rangle e_\alpha \right\rangle \\ &= \langle ue_i, ue_j \rangle = \langle e_i, e_j \rangle = \delta_{ij}. \end{aligned}$$

Therefore, we have

$$dW_t^j = \langle e_\alpha, U_t e_j \rangle dM_t^\alpha,$$

which implies that W is a continuous $\mathbb{F}^X$ -local martingale since M^α are $\mathbb{F}^X$ -local martingales. Again, because

$$d\langle M^\alpha, M^\beta \rangle_s = \langle P_\alpha, P_\beta \rangle ds,$$

we have

$$\begin{aligned} \langle W_t^i, W_t^j \rangle &= \sum_{\alpha, \beta} \int_0^t \langle e_\alpha, U_s e_j \rangle \langle e_\beta, U_s e_j \rangle \langle P_\alpha, P_\beta \rangle ds \\ &= \sum_{\alpha} \int_0^t \langle e_\alpha, U_s e_j \rangle \left\langle \sum_{\beta} \langle P_\alpha, P_\beta \rangle e_\beta, U_s e_j \right\rangle ds \\ &= \sum_{\alpha} \int_0^t \langle e_\alpha, U_s e_j \rangle \left\langle \sum_{\beta} \langle P P_\alpha, e_\beta \rangle e_\beta, U_s e_j \right\rangle ds \\ &= \sum_{\alpha} \int_0^t \langle e_\alpha, U_s e_j \rangle \langle P_\alpha, U_s e_j \rangle ds \\ &= \int_0^t \sum_{\alpha} \langle e_\alpha, U_s e_j \rangle \langle e_\alpha, U_s e_j \rangle ds = \delta^{ij} t. \end{aligned}$$

Therefore, by the Lévy's theorem, W is a $\mathbb{F}^X$ -Brownian motion. $\square$

Remark 3.2.3. We can make $\mathbb{F}^X$ larger. X is called a $\mathbb{F}$ -Brownian motion if it is defined as above w.s.t. $\mathbb{F}$ and it is a strong Markov process w.s.t. $\mathbb{F}$.

If X be a Brownian motion on M , then its horizontal lift U is called a horizontal Brownian motion. Since

$$dU_t = H_i(U_t) \circ dW_t^i,$$

where W is a Brownian motion, then U is a diffusion process generated by the operator

$$L = \frac{1}{2} \sum_i H_i^2 = \frac{1}{2} \Delta_{O(M)}.$$

Proposition 3.2.4. *Let $U: \Omega \rightarrow W(O(M))$ be a measurable map defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. TFAE.*

- (1) U is a horizontal $\mathbb{F}^U$ -semimartingale whose projection $X = \pi(U)$ is a Brownian motion on M .
- (2) U is a horizontal $\mathbb{F}^U$ -semimartingale whose anti-development is a standard Euclidean Brownian motion.
- (3) U is a $\Delta_{O(M)}/2$ -diffusion w.s.t. $\mathbb{F}^U$.

Remark 3.2.5. An $O(M)$ -valued process U satisfies any of above is called a horizontal Brownian motion on $O(M)$.

Proof. (1) $\Rightarrow$ (2) : Since X is a Brownian motion, by above its anti-development W is a standard Euclidean Brownian motion.

(2) $\Rightarrow$ (3) : Then by

$$dU_t = H_i(U_t) \circ dW_t^i,$$

it is a diffusion process of $\Delta_{O(M)}/2$.

(3) $\Rightarrow$ (1) : For any $f \in C^\infty(M)$, let $\tilde{f} = f \circ \pi \in C^\infty(O(M))$. Because

$$\Delta_M f(\pi(u)) = \Delta_{O(M)} \tilde{f}(u),$$

we have

$$f(X_t) - f(X_0) - \frac{1}{2} \int_0^t \Delta_M f(X_s) ds = \tilde{f}(U_t) - \tilde{f}(U_0) - \frac{1}{2} \int_0^t \Delta_{O(M)} \tilde{f}(U_s) ds.$$

The RHS is a $\mathbb{F}^U$ -local martingale, which means that the LHS is also a $\mathbb{F}^U$ -local martingale. Moreover, $X = \pi(U)$ implies $\mathbb{F}^X \subset \mathbb{F}^U$, so the LHS is a $\mathbb{F}^X$ -local martingale. $\square$

Remark 3.2.6. Note that let $M \subset \mathbb{R}^N$ and B is a Brownian motion on $\mathbb{R}^N$. Then if

$$dX_t = P_\alpha(X_t) \circ dB_t^\alpha,$$

then X_t is a diffusion process generated by

$$L = \frac{1}{2} \sum_{\alpha} P_\alpha^2 = \frac{1}{2} \Delta_M.$$

Therefore, X is a Brownian motion on M .

3.3 Examples of Brownian Motion

Riemannian Structure of Sphere. First, consider the standard Riemannian structure on $\mathbb{S}^d$.

1. $d = 1$: Choosing a chart $(0, 2\pi) \rightarrow \mathbb{S}^1$ and consider the embedding

$$\iota: \mathbb{S}^1 \hookrightarrow \mathbb{R}^2$$

given by $\theta \mapsto (x, y)$, where $x = \cos \theta$ and $y = \sin \theta$. Note that any $V_{(x,y)} \in T_{(x,y)} \mathbb{R}^2$ is equivalent to $V_{(x,y)} \in \mathbb{R}^2$ by setting

$$V_{(x,y)} = a(x, y) \frac{\partial}{\partial x} + b(x, y) \frac{\partial}{\partial y} = (a(x, y), b(x, y))^\top.$$

and for any $f(x, y) \in C^\infty(\mathbb{R}^2)$,

$$V_{(x,y)} f = a(x, y) \frac{\partial f}{\partial x}(x, y) + b(x, y) \frac{\partial f}{\partial y}(x, y).$$

So for $\iota_*: T\mathbb{S}^1 \rightarrow \mathbb{R}^2$, at θ_0

$$\iota_{*, \theta_0} \left(\frac{d}{d\theta} \Big|_{\theta_0} \right) = -\sin \theta_0 \frac{\partial}{\partial x} + \cos \theta_0 \frac{\partial}{\partial y}.$$

which is because for any $f(x, y) \in C^\infty(M)$,

$$\iota_{*, \theta_0} \left(\frac{d}{d\theta} \Big|_{\theta_0} \right) f = \frac{d}{d\theta} \Big|_{\theta_0} f \circ \iota = \frac{d}{d\theta} \Big|_{\theta_0} f(\cos \theta, \sin \theta) = -\sin \theta_0 \frac{\partial f}{\partial x}(\theta_0) + \cos \theta_0 \frac{\partial f}{\partial y}(\theta_0).$$

And note that ι_* is $C^\infty(\mathbb{S}^1)$ -linear. Then the induce metric on $\mathbb{S}^1$ is given by

$$g_{\theta\theta}(\theta_0) = \left\langle \frac{d}{d\theta} \Big|_{\theta_0}, \frac{d}{d\theta} \Big|_{\theta_0} \right\rangle = \left\langle \iota_{*,\theta_0} \frac{d}{d\theta} \Big|_{\theta_0}, \iota_{*,\theta_0} \frac{d}{d\theta} \Big|_{\theta_0} \right\rangle = \langle (-\sin \theta_0, \cos \theta_0), (-\sin \theta_0, \cos \theta_0) \rangle = 1$$

Therefore, $g = d\theta \otimes d\theta$, which implies that the corresponding Christoffel coefficients

$$\Gamma_{\theta\theta}^\theta = \frac{1}{2} g^{\theta\theta} \frac{d}{d\theta} g_{\theta\theta} = 0,$$

It follows that that for any $X = a(\theta) \frac{d}{d\theta}, Y = b(\theta) \frac{d}{d\theta}$, we have

$$\nabla_X Y = a(\theta) \dot{b}(\theta) \frac{d}{d\theta},$$

because $\nabla_{\frac{d}{d\theta}} \frac{d}{d\theta} = 0$. Moreover, we also have

$$\Delta_{\mathbb{S}^1} h(\theta) = \frac{d^2}{d\theta^2} h \Rightarrow \Delta_{\mathbb{S}^1} = \frac{d^2}{d\theta^2}.$$

On the other hand, consider $\mathbb{S}^1 \subset \mathbb{R}^2$ embedded, let $T = \iota_*(d/d\theta)$. Then

$$\tilde{X} = \iota_*(X) = a(\theta)T, \quad \tilde{Y} = \iota_*(Y) = b(\theta)T.$$

Moreover, they can be extended to vector fields on $\mathbb{R}^2$. Let $\tilde{T} = (-y, x)$, i.e., $T = \tilde{T}|_{\mathbb{S}^1}$. Then

$$\tilde{X}(x, y) = \tilde{a}(x, y)\tilde{T}, \quad \tilde{Y}(x, y) = \tilde{b}(x, y)\tilde{T},$$

where $\tilde{a}(\cos \theta, \sin \theta) = a(\theta)$ and $\tilde{b}(\cos \theta, \sin \theta) = b(\theta)$. Note that

$$\begin{aligned} \tilde{T}(\tilde{a})|_{\mathbb{S}^1}(\theta) &= -y \frac{\partial \tilde{a}}{\partial x} + x \frac{\partial \tilde{a}}{\partial y} = \dot{a}(\theta) \\ \tilde{T}(\tilde{b})|_{\mathbb{S}^1}(\theta) &= -y \frac{\partial \tilde{b}}{\partial x} + x \frac{\partial \tilde{b}}{\partial y} = \dot{b}(\theta) \end{aligned}$$

and so we have

$$\tilde{\nabla}_{\tilde{T}} \tilde{T}|_{\mathbb{S}^1}(\theta) = -\tilde{T}(y) \frac{\partial}{\partial x} + \tilde{T}(x) \frac{\partial}{\partial y} = -\cos \theta \frac{\partial}{\partial x} - \sin \theta \frac{\partial}{\partial y},$$

which mean that $\tilde{\nabla}_{\tilde{T}} \tilde{T}|_{\mathbb{S}^1} = -\vec{n} \in N_\theta \mathbb{S}^1$. So

$$\nabla_T T = (\tilde{\nabla}_{\tilde{T}} \tilde{T})^\top = 0,$$

and by viewing $T = d/d\theta$, $\Gamma_{\theta\theta}^\theta = 0$ as same calculation as above. Moreover,

$$\tilde{\nabla}_{\tilde{X}} \tilde{Y}|_{\mathbb{S}^1}(\theta) = a(\theta) \left(\tilde{T}(\tilde{b})\tilde{T} + \tilde{b} \tilde{\nabla}_{\tilde{T}} \tilde{T} \right) = a(\theta) \dot{b}(\theta) T - a(\theta) b(\theta) \vec{n}(\theta).$$

It follows that

$$\nabla_X Y(\theta) = (\tilde{\nabla}_{\tilde{X}} \tilde{Y})^\top = a(\theta) \dot{b}(\theta) \frac{d}{d\theta},$$

and

$$\Pi(X, Y) = (\tilde{\nabla}_{\tilde{X}} \tilde{Y})^\perp = -a(\theta) b(\theta) \vec{n}(\theta) = -\langle X, Y \rangle \vec{n}.$$

2. In general case, by embedding $\mathbb{S}^d \subset \mathbb{R}^{d+1}$,

$$\nabla_X Y = \tilde{\nabla}_X Y + \langle X, Y \rangle \vec{n}$$

where $\vec{n} \in \Gamma(N\mathbb{S}^d)$ is the outer unit normal vector. Then let $\{X_i\}_{i=1}^d$ be an orthonormal frame of $T\mathbb{S}^d$ and so $\{X_i, \vec{n}\}$ is an orthonormal frame of $\mathbb{R}^{d+1}$. For any $f \in C^\infty(\mathbb{S}^d)$, let $F \in C^\infty(\mathbb{R}^{d+1})$ be an extension of f . As we know $\text{grad } f = (\overline{\text{grad } f})^\top$, $\tilde{\nabla}_{X_i} \vec{n} = X_i$, and

$$\Delta_{\mathbb{R}^{d+1}} F = \sum_{i=1}^d \tilde{\nabla}^2 F(X_i, X_i) + \tilde{\nabla}^2 F(\vec{n}, \vec{n}),$$

we have

$$\begin{aligned} \Delta_{\mathbb{S}^d} f &= \sum_{i=1}^d \nabla^2 f(X_i, X_i) = \sum_{i=1}^d \langle \nabla_{X_i} \text{grad } f, X_i \rangle \\ &= \sum_{i=1}^d \left\langle \tilde{\nabla}_{X_i} \text{grad } F + \langle \text{grad } F, X_i \rangle \vec{n}, X_i \right\rangle = \sum_{i=1}^d \left\langle \tilde{\nabla}_{X_i} \text{grad } F, X_i \right\rangle \\ &= \sum_{i=1}^d \left\langle \tilde{\nabla}_{X_i} (\overline{\text{grad } F})^\top, X_i \right\rangle = \sum_{i=1}^d \left\langle \tilde{\nabla}_{X_i} (\overline{\text{grad } F} - \langle \overline{\text{grad } F}, \vec{n} \rangle \vec{n}), X_i \right\rangle \\ &= \sum_{i=1}^d \left\langle \tilde{\nabla}_{X_i} \overline{\text{grad } F}, X_i \right\rangle - d \langle \overline{\text{grad } F}, \vec{n} \rangle = \sum_{i=1}^d \tilde{\nabla}^2 F(X_i, X_i) - d \tilde{\nabla}_{\vec{n}} F \\ &= \Delta_{\mathbb{R}^{d+1}} F - \tilde{\nabla}^2 F(\vec{n}, \vec{n}) - d \tilde{\nabla}_{\vec{n}} F \end{aligned}$$

On the other hand, choose a polar coordinate chart for $\mathbb{R}^{d+1} \setminus \{0\}$, $(r, \theta = (\theta^1, \dots, \theta^d))$. Then for Euclidean coordinate $(x^1, \dots, x^{d+1})$,

$$x^i = r \iota^i(\theta),$$

where $\iota: \mathbb{S}^d \hookrightarrow \mathbb{R}^{d+1}$ is the inclusion. Then we have

$$\frac{\partial}{\partial r} = \frac{\partial x^i}{\partial r} \frac{\partial}{\partial x^i} = \iota^i \frac{\partial}{\partial x^i}$$

and

$$\frac{\partial}{\partial \theta^\alpha} = \frac{\partial x^i}{\partial \theta^\alpha} \frac{\partial}{\partial x^i} = r \frac{\partial \iota^i}{\partial \theta^\alpha} \frac{\partial}{\partial x^i}$$

Then we have, first

$$g_{rr} = \left\langle \frac{\partial}{\partial r}, \frac{\partial}{\partial r} \right\rangle = \delta_{ij} \iota^i \iota^j = 1,$$

which also implies that

$$0 = \frac{\partial}{\partial \theta^\alpha} \delta_{ij} \iota^i \iota^j = 2 \delta_{ij} \iota^i \frac{\partial \iota^j}{\partial \theta^\alpha} = 0.$$

So, second

$$g_{\alpha r} = \left\langle \frac{\partial}{\partial \theta^\alpha}, \frac{\partial}{\partial r} \right\rangle = r \delta_{ij} \iota^i \frac{\partial \iota^j}{\partial \theta^\alpha} = 0.$$

Finally,

$$g_{\alpha\beta} = \left\langle \frac{\partial}{\partial \theta^\alpha}, \frac{\partial}{\partial \theta^\beta} \right\rangle = r^2 \delta_{ij} \frac{\partial \iota^i}{\partial \theta^\alpha} \frac{\partial \iota^j}{\partial \theta^\beta}.$$

For $g_{\alpha\beta}$, note that ι is the isometric inclusion, so

$$g_{\mathbb{S}^d} \left(\frac{\partial}{\partial \theta^\alpha} \Big|_{\mathbb{S}^d}, \frac{\partial}{\partial \theta^\beta} \Big|_{\mathbb{S}^d} \right) = g \left(\iota_* \frac{\partial}{\partial \theta^\alpha} \Big|_{\mathbb{S}^d}, \iota_* \frac{\partial}{\partial \theta^\beta} \Big|_{\mathbb{S}^d} \right) = \delta_{ij} \frac{\partial \iota^i}{\partial \theta^\alpha} \frac{\partial \iota^i}{\partial \theta^\beta},$$

because

$$\iota_* \frac{\partial}{\partial \theta^\alpha} \Big|_{\mathbb{S}^d} = \frac{\partial \iota^i}{\partial \theta^\alpha} \frac{\partial}{\partial x^i}.$$

It follows that

$$g_{\alpha\beta} = r^2 h_{\alpha\beta}, \quad h_{\alpha\beta} := g_{\mathbb{S}^d} \left(\frac{\partial}{\partial \theta^\alpha} \Big|_{\mathbb{S}^d}, \frac{\partial}{\partial \theta^\beta} \Big|_{\mathbb{S}^d} \right).$$

So we have

$$g = dr \otimes dr + r^2 h_{\alpha\beta} d\theta^\alpha \otimes d\theta^\beta.$$

It also implies that (r, θ) is a semi-geodesic coordinate of $\mathbb{R}^{d+1}$ w.s.t. $\mathbb{S}^d$ and distance function r . Moreover, based on this coordinate system, we have

$$g = (g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & r^2 h_{\alpha\beta} \end{pmatrix}, \quad g^{-1} = (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & r^{-2} h^{\alpha\beta} \end{pmatrix},$$

which implies that

$$\sqrt{|g|} = r^d \sqrt{|h|}.$$

Let $(y^1, \dots, y^{d+1}) = (r, \theta^1, \dots, \theta^d)$. For any $F = F(r, \theta) \in C^\infty(\mathbb{R}^{d+1})$, note that

$$g^{ij} \frac{\partial F}{\partial y^j} = \begin{cases} \frac{\partial F}{\partial r}, & i = 1 \\ \frac{h^{i\beta}}{r^2} \frac{\partial F}{\partial \theta^\beta}, & i \neq 1 \end{cases}$$

Then by the definition of Laplace-Beltrami operator,

$$\begin{aligned} \Delta_{\mathbb{R}^{d+1}} F &= \frac{1}{|g|} \frac{\partial}{\partial y^i} \left(\sqrt{|g|} g^{ij} \frac{\partial F}{\partial y^j} \right) \\ &= \frac{1}{|g|} \frac{\partial}{\partial r} \left(\sqrt{|g|} \frac{\partial F}{\partial r} \right) + \frac{1}{|g|} \frac{\partial}{\partial \theta^\alpha} \left(\sqrt{|g|} \frac{h^{\alpha\beta}}{r^2} \frac{\partial F}{\partial \theta^\beta} \right) = \text{I} + \text{II}. \end{aligned}$$

For I,

$$\begin{aligned} \text{I} &= \frac{1}{|g|} \frac{\partial}{\partial r} \left(\sqrt{|g|} \frac{\partial F}{\partial r} \right) = \frac{1}{r^d \sqrt{|h|}} \frac{\partial}{\partial r} \left(r^d \sqrt{|h|} \frac{\partial F}{\partial r} \right) \\ &= \frac{d}{r} \frac{\partial F}{\partial r} + \frac{\partial^2 F}{\partial r^2}. \end{aligned}$$

For II,

$$\begin{aligned} \text{II} &= \frac{1}{|g|} \frac{\partial}{\partial \theta^\alpha} \left(\sqrt{|g|} \frac{h^{\alpha\beta}}{r^2} \frac{\partial F}{\partial \theta^\beta} \right) \\ &= \frac{1}{r^d \sqrt{|h|}} \frac{\partial}{\partial \theta^\alpha} \left(r^{d-2} \sqrt{|h|} h^{\alpha\beta} \frac{\partial F}{\partial \theta^\beta} \right) \\ &= \frac{1}{r^2 \sqrt{|h|}} \frac{\partial}{\partial \theta^\alpha} \left(\sqrt{|h|} h^{\alpha\beta} \frac{\partial F}{\partial \theta^\beta} \right) \\ &= \frac{1}{r^2} \Delta_{\mathbb{S}^d} \hat{F}_r, \end{aligned}$$

where $\hat{F}_r(\theta) = F(r, \theta) \in C^\infty(\mathbb{S}^d)$. So we have

$$\Delta_{\mathbb{R}^{d+1}} F = \frac{d}{r} \frac{\partial F}{\partial r} + \frac{\partial^2 F}{\partial r^2} + \frac{1}{r^2} \Delta_{\mathbb{S}^d} \hat{F}_r.$$

Then on $\mathbb{S}^d$, i.e., $r = 1$, for any $f \in C^\infty(\mathbb{S}^d)$, choose any extension F , then

$$\Delta_{\mathbb{S}^d} f = \left(\Delta_{\mathbb{R}^{d+1}} - d \frac{\partial}{\partial r} - \frac{\partial^2}{\partial r^2} \right) F|_{r=1}.$$

In particular, if we choose $F(r, \theta) = f(\theta)$ for any r , then

$$\Delta_{\mathbb{S}^d} f = \Delta_{\mathbb{R}^{d+1}} F.$$

Remark 3.3.1. If we embedded $\iota: \mathbb{S}^d \hookrightarrow \mathbb{R}^{d+1}$ by

$$\begin{aligned} x_1 &= \cos \theta_1 \\ x_2 &= \sin \theta_1 \cos \theta_2 \\ x_3 &= \sin \theta_1 \sin \theta_2 \cos \theta_3 \\ &\vdots \\ x_d &= \sin \theta_1 \cdots \sin \theta_{d-1} \cos \theta_d \\ x_{d+1} &= \sin \theta_1 \cdots \sin \theta_{d-1} \sin \theta_d \end{aligned}$$

then the induced metric metric is given by

$$g_{\mathbb{S}^d} = d\theta_1 \otimes d\theta_1 + \sin^2 \theta_1 d\theta_2 \otimes d\theta_2 + \sin^2 \theta_1 \sin^2 \theta_2 d\theta_3 \otimes d\theta_3 + \cdots + \left(\prod_{i=1}^{d-1} \sin^2 \theta_i \right) d\theta_d \otimes d\theta_d.$$

Note that it implies that $\left\{ \frac{\partial}{\partial \theta^\alpha} \right\}$ is a orthogonal frame of $\mathbb{S}^d$ but not normal.

Radially Symmetric Space. For a Riemannian manifold (M, g) of dimension d , consider a normal coordinate system, i.e., $\exp_p: U \rightarrow M$, and using the polar coordinate $(r, \theta) \in (0, \infty) \times \mathbb{S}^{d-1}$, then we have

$$g = dr \otimes dr + g_{\alpha\beta}(r, \theta) d\theta^\alpha \otimes d\theta^\beta.$$

Definition 3.3.2. A Riemannian manifold (M, g) is called radially symmetric if on any normal polar chart, for

$$g = dr \otimes dr + g_{\alpha\beta}(r, \theta) d\theta^\alpha \otimes d\theta^\beta,$$

it has

$$g_{\alpha\beta}(r, \theta) = G(r)^2 h_{\alpha\beta}(\theta),$$

where

$$G(0) = 0, \quad G'(0) = 1,$$

and $h_{\alpha\beta} d\theta^\alpha \otimes d\theta^\beta$ is the canonical Riemannian metric on $\mathbb{S}^{d-1}$.

Proposition 3.3.3. Let (M, g) be a Riemannian manifold with constant sectional curvature κ . Then M is radially symmetric, i.e.,

$$g = dr \otimes dr + G(r)^2 h_{\alpha\beta}(\theta) d\theta^\alpha \otimes d\theta^\beta,$$

where $G(0) = 0$, $G'(0) = 1$. Moreover,

$$G''(r) + \kappa G(r) = 0.$$

Proof. Fix $p \in M$, and on $T_p M$, choosing an orthonormal basis such that $T_p M = \mathbb{R}^d$. For the $\Phi: (0, \delta) \times \mathbb{S}^{d-1} \rightarrow M$, $\Phi(r, \theta) = \exp_p(r\theta)$ is a geodesic, which also provides the normal polar coordination (r, θ) . Fix a $\theta_0 \in \mathbb{S}^{d-1}$. Consider a normal radial geodesic $\gamma(r)$ with $\gamma(0) = p$ and $\dot{\gamma}(0) = \theta_0 \in \mathbb{S}^{d-1}$. Let $\theta_\alpha: (-\varepsilon, \varepsilon) \rightarrow \mathbb{S}^{d-1}$ be a smooth map with $\theta(0) = \theta_0$ and $\dot{\theta}(0) = \frac{\partial}{\partial \theta^\alpha}|_{\mathbb{S}^{d-1}, \theta_0}$. Note that:

- (i) $W_\alpha = \frac{\partial}{\partial \theta^\alpha}|_{\mathbb{S}^{d-1}, \theta_0} = \iota_* \frac{\partial}{\partial \theta^\alpha}|_{\mathbb{S}^{d-1}, \theta_0} \in \mathbb{R}^d$, i.e., an vector in $T_p M$,
- (ii) because $|\theta(s)| \equiv 1$, $W_\alpha \perp \theta_0$.

Consider the radial variation

$$F_\alpha(s, r) = \exp_p(r\theta_\alpha(s))$$

and its two vector fields

$$V_\alpha(r) = \frac{\partial F_\alpha}{\partial s}(r, 0) = (d \exp_p)_{r\theta_0}(rW_\alpha), \quad T(r) = \frac{\partial F_\alpha}{\partial r}(r, 0) = \dot{\gamma}(r).$$

Then $V_\alpha(r)$ is a Jacobian field. Moreover, we have

$$V_\alpha(0) = 0, \quad \nabla_T V_\alpha(0) = W_\alpha.$$

Because $W_\alpha \perp T(0) = \theta_0$ and $V_\alpha(0) \perp T(0)$, $V_\alpha \perp T$. It implies that

$$R(V_\alpha, T)T = \kappa V_\alpha,$$

since M has constant curvature, i.e.

$$\langle R(V_\alpha, T)T, V_\alpha \rangle = \kappa (|V_\alpha|^2 |T|^2 - \langle V_\alpha, T \rangle^2) = \kappa |V_\alpha|^2.$$

So the Jacobian equation is

$$\frac{D^2}{dt^2} V_\alpha(t) + \kappa V_\alpha = 0.$$

So when choosing an orthogonal frame $\{E_i(r)\}$ with $E_1(r) = T(r)$ along $\gamma(r)$ and $E_\alpha(r)$ are obtained by parallel moving W_α along $\gamma(r)$, which are orthogonal because $\langle W_\alpha, W_\beta \rangle = 0$. Using similar result, because

$$\langle V_\alpha(0), E_\beta(0) \rangle = \langle \nabla_T V_\alpha(0), E_\beta(0) \rangle = \langle W_\alpha, W_\beta \rangle = 0, \quad \forall \beta \neq \alpha,$$

$V_\alpha \perp E_\beta$. So

$$V_\alpha(r) = G_\alpha(r)E_\alpha(r)$$

and $G_\alpha(r)$ should satisfy

$$f''(r) + \kappa f(r) = 0, \quad f(0) = 0, \quad f'(0) = 1$$

By the uniqueness of ODE, for all α

$$G(r) = G_\alpha(r) = \begin{cases} \frac{1}{\sqrt{\kappa}} \sin(\sqrt{\kappa}r), & \kappa > 0, \\ r, & \kappa = 0, \\ \frac{1}{\sqrt{-\kappa}} \sinh(\sqrt{-\kappa}r), & \kappa < 0, \end{cases}$$

Next, the the coordinate tangent vectors on M are given by

$$\frac{\partial}{\partial \theta^\alpha} \Big|_M = \Phi_* \left(\frac{\partial}{\partial \theta^\alpha} \Big|_{\mathbb{S}^{d-1}} \right)$$

So by definition

$$d\Phi_{(r,\theta_0)} \frac{\partial}{\partial \theta^\alpha} \Big|_{\mathbb{S}^{d-1}, \theta_0} = \frac{d}{ds} \Big|_{s=0} \Phi(r, \theta_\alpha(s)) = \frac{d}{ds} \Big|_{s=0} \exp_p(r\theta_\alpha(s)) = V_\alpha(r),$$

i.e.,

$$\frac{\partial}{\partial \theta^\alpha} \Big|_{M, (r, \theta_0)} = V_\alpha(r).$$

Therefore,

$$g_{\alpha\beta}(r, \theta_0) = \langle V_\alpha(r), V_\beta(r) \rangle = G(r)^2 \langle E_\alpha(r), E_\beta(r) \rangle = G(r)^2 \langle E_\alpha(0), E_\beta(0) \rangle = G(r)^2 \langle W_\alpha, W_\beta \rangle$$

because $E_\alpha(r)$ are obtained by parallel moving. Let

$$h_{\alpha\beta}(\theta_0) = \langle W_\alpha, W_\beta \rangle = \left\langle \frac{\partial}{\partial \theta^\alpha} \Big|_{\mathbb{S}^{d-1}, \theta_0}, \frac{\partial}{\partial \theta^\beta} \Big|_{\mathbb{S}^{d-1}, \theta_0} \right\rangle,$$

i.e., $h_{\alpha\beta} d\theta^\alpha \otimes d\theta^\beta$ is the canonical Riemannian metric on $\mathbb{S}^{d-1}$. So we have

$$g = dr \otimes dr + G(r)^2 h_{\alpha\beta}(\theta) d\theta^\alpha \otimes d\theta^\beta. \quad \square$$

For a radially symmetric space (M, g) , using similar calculation as $\mathbb{S}^d$, we have

$$\Delta_M = L_r + \frac{1}{G(r)^2} \Delta_{\mathbb{S}^{d-1}},$$

where

$$L_r = \frac{\partial^2}{\partial r^2} + (d-1) \frac{G'(r)}{G(r)} \frac{\partial}{\partial r},$$

is called the radial Laplacian.

Examples.

Example 3.3.4 (Circle). Consider

$$\mathbb{S}^1 = \{e^{i\theta} : 0 \leq \theta < 2\pi\} \subset \mathbb{R}^2.$$

Let B be a Brownian motion. Define

$$X_t = e^{iB_t} = (\cos B_t, \sin B_t).$$

Then X is a Brownian motion on $\mathbb{S}^1$.

Proof by Laplacian Operator. First, When viewing $\mathbb{S}^1 \subset \mathbb{R}^2$, $X_t = (\cos B_t, \sin B_t)$, by Itô formula,

$$dX_t = (-\sin B_t, \cos B_t) \circ dB_t = \iota_* \frac{d}{d\theta}(B_t) \circ dB_t.$$

Therefore,

$$V = \iota_* \frac{d}{d\theta} = \frac{d}{d\theta}.$$

Then X_t is a diffusion process with

$$L = \frac{1}{2} \frac{d^2}{d\theta^2} = \frac{1}{2} \Delta_{\mathbb{S}^1}.$$

So X is a Brownian motion on $\mathbb{S}^1$. □

Proof by Anti-development. First, note that at any $\theta \in \mathbb{S}^k$, $O(\mathbb{S}^1)_\theta = \{-1, 1\}$ and so the orthonormal frame bundle $O(M) = \bigcup_{\theta \in \mathbb{S}^1} \{-1, 1\}$. By $\Gamma_{\theta\theta}^\theta = 0$, any horizontal vector $\dot{u}(0)$ vanishes on the vertical part. Then by $dU_t = P_\alpha^H \circ dX_t^\alpha$, the horizontal lift of X starting from $U_0 = 1$, $U_t \equiv 1$.

On the other hand, consider $P(\theta): \mathbb{R}^2 \rightarrow T_\theta \mathbb{S}^1$. Note that $n = (-\sin \theta, \cos \theta) \in N_\theta \mathbb{S}^1$. So

$$P(\theta) = \begin{pmatrix} \sin^2 \theta & -\sin \theta \cos \theta \\ -\sin \theta \cos \theta & \cos^2 \theta \end{pmatrix}.$$

Therefore,

$$\begin{aligned} P_1(\theta) &= (\sin^2 \theta, -\sin \theta \cos \theta)^\top = \iota_* \left(-\sin \theta \frac{d}{d\theta} \right) \\ P_2(\theta) &= (-\sin \theta \cos \theta, \cos^2 \theta)^\top = \iota_* \left(\cos \theta \frac{d}{d\theta} \right). \end{aligned}$$

Then by $dW^t = U_t^{-1} P_\alpha(X_t) \circ dX_t^\alpha$,

$$\begin{aligned} dW_t &= -\sin B_t \circ dX_t^1 + \cos B_t \circ dX_t^2 \\ &= \sin^2 B_t \circ dB_t + \cos^2 B_t \circ dB_t \\ &= dB_t, \end{aligned}$$

and $W_0 = 0$, so $W_t = B_t$. □

Example 3.3.5 (Sphere). Consider $\mathbb{S}^d \subset \mathbb{R}^{d+1}$. Let $\vec{n}$ be the outer unit normal vector. Then the orthogonal projection

$$P(\theta): \mathbb{R}^{d+1} \rightarrow T_\theta \mathbb{S}^d$$

is given by $P(\theta)(x) = x - \langle x, \vec{n}(\theta) \rangle \vec{n}(\theta)$. The matrix expression of $P(\theta)$ is

$$P(\theta)_{ij} = \delta_{ij} - n(\theta)_i n(\theta)_j$$

Or when embedding $\mathbb{S}^d \hookrightarrow \mathbb{R}^{d+1}$, $P(x)_{ij} = \delta_{ij} - x^i x^j$. Therefore, by $\Delta_M = \sum_\alpha P_\alpha^2$, X is a Brownian motion on $\mathbb{S}^d$, when it satisfies

$$X_t^i = X_0^i + \sum_{j=1}^{d+1} \int_0^t (\delta_{ij} - X_s^i X_s^j) \circ dW_s^j,$$

where W is a standard Brownian motion on $\mathbb{R}^{d+1}$.

Example 3.3.6 (Radially Symmetric Space). Let (M, g) be a radially symmetric space and $X_t = (r_t, \theta_t)$ be a Brownian motion on M expressed as polar coordinate. Consider its radial process and angular process separately.

(1) Radial process: Choose $f(r, \theta) = r$ to the martingale problem,

$$\begin{aligned} \beta_t &= r_t - r_0 - \frac{1}{2} \int_0^t \Delta_M f(X_s) ds \\ &= r_t - r_0 - \frac{d-1}{2} \int_0^t \frac{G'(r_s)}{G(r_s)} ds \end{aligned}$$

is a local martingale. Moreover, the quadratic variation is given by

$$\langle \beta, \beta \rangle_t = \int_0^t \Gamma(r, r)(X_s) ds,$$

where we have

$$\Gamma(r, r) = \langle \nabla r, \nabla r \rangle = g^{ij} \frac{\partial r}{\partial x^i} \frac{\partial r}{\partial x^j} = |\nabla r|^2 = 1.$$

So

$$\langle \beta, \beta \rangle_t = t,$$

which implies that β is a 1-dimensional Brownian motion. On the other hand, we can see the radial process r_t is generated by the radial operator L_r .

- (2) Angular process: Let $f \in C^\infty(\mathbb{S}^{d-1})$, which can be extended to $f(r, \theta) = f(\theta) \in C^\infty(M)$. Then

$$f(X_t) = f(\theta_t), \quad \Delta_M f(r, \theta) = \frac{1}{G(r)^2} \Delta_{\mathbb{S}^{d-1}} f.$$

It implies that

$$M_t^f = f(\theta_t) - f(\theta_0) - \frac{1}{2} \int_0^t \frac{1}{G(r_s)^2} \Delta_{\mathbb{S}^{d-1}} f(\theta_s) ds$$

Let

$$T_{t'} = \int_0^{t'} \frac{1}{G(r_s)^2} ds,$$

which is a nondecreasing process. So we can consider

$$\tau_t = \inf \{t' \geq 0 : T_{t'} \geq t\} = T^{-1}(t),$$

where the final equality is because T is strictly increasing in t . Then by considering the time-change, let $z_t = \theta_{\tau_t}$, we have

$$M_{\tau_t}^f = f(z_t) - f(z_0) - \frac{1}{2} \int_0^t \Delta_{\mathbb{S}^{d-1}} f(z_s) ds.$$

Since M_t is a local martingale, $t \mapsto M_{\tau_t}^f$ is also a local martingale w.s.t. $\mathcal{F}_{\tau_t}$. Therefore, $t \mapsto z_t = \theta_{\tau_t}$ is a Brownian motion on $\mathbb{S}^{d-1}$.

Claim: r and z are independent.

Proof. I. First, we embed $\mathbb{S}^{d-1} \hookrightarrow \mathbb{R}^d$ with $\theta \mapsto y = (y^\alpha)$ and choose $f(\theta) = f(y) = f^\alpha(y) = y^\alpha$, the coordinate function on $\mathbb{R}^\alpha$. Let U_t be a horizontal lift of $\theta_t = Y_t \in \mathbb{R}^d$ and W_t be its anti-development. Then

$$W_t = \int_0^t U_s^{-1} P_\alpha(Y_s) \circ dY_s^\alpha.$$

Because τ_t is continuous, for any continuous semimartingales H, K ,

$$\int_0^{\tau_t} H_s dK_s = \int_0^t H_{\tau_s} dK_{\tau_s}, \quad \langle H \circ \tau, K \circ \tau \rangle_t = \langle H, K \rangle_{\tau_t}.$$

So U_{τ_t} is a horizontal lift of θ_{τ_t} and W_{τ_t} is the anti-development of θ_{τ_t} . Because θ_{τ_t} is a Brownian motion on $\mathbb{S}^{d-1}$, $\{W_{\tau_t}\}_{t \geq 0}$ is an Euclidean Brownian motion.

II. Next, we need to check

$$\langle W \circ \tau, \beta \rangle = 0$$

For any $f \in C^\infty(\mathbb{S}^{d-1})$,

$$\langle M^f, \beta \rangle_t = \int_0^t \Gamma(f, r)(X_s) ds = \int_0^t \langle \nabla f, \nabla r \rangle(X_s) ds.$$

Note that f is viewing as $f(r, \theta) = f(\theta) \in C^\infty(M)$. By

$$\langle \nabla f^\alpha, \nabla r \rangle = g^{ij} \frac{\partial f^\alpha}{\partial x^i} \frac{\partial r}{\partial x^j} = 0,$$

we have

$$\langle M^{f^\alpha}, \beta \rangle = \langle Y^\alpha, \beta \rangle \equiv 0, \quad \forall \alpha = 1, 2, \dots, d,$$

which implies that

$$\langle W, \beta \rangle_t = \int_0^t U_s^{-1} P_\alpha(Y_s) d\langle Y^\alpha, \beta \rangle = 0$$

By the following lemma, we have

$$\langle W \circ \tau, \beta \rangle_t = 0.$$

III. Because $\{W_{\tau_t}\}_{t \geq 0}$ and $\{\beta_t\}_{t \geq 0}$ are two Brownian motions, they are independent. Moreover, since

$$d\theta_{\tau_t} = U_{\tau_t}^{-1} e_i \circ dW_{\tau_t}^i$$

and

$$dr_t = d\beta_t + \frac{d-1}{2} \frac{G'(r_t)}{G(r_t)} dt,$$

r_t and $z_t = \theta_{\tau_t}$ are independent. □

Lemma 3.3.7. *Let W and M be two continuous local martingale and τ be a continuous time-change. If $\langle W, M \rangle \equiv 0$, then $\langle W \circ \tau, M \rangle \equiv 0$.*

Example 3.3.8 (In Local Coordinates). Let $(x^1, \dots, x^d)$ be a local chart of M and $(x^i, i = 1, \dots, d; e_j^i, 1 \leq i \leq j \leq d)$ be a chart of $O(M)$. The equation of a horizontal Brownian motion is

$$dU_t = H_i(U_t) \circ dW_t^i,$$

where W is a d -dimensional Brownian motion. We have already shown that

$$H_i(u) = e_j^i \frac{\partial}{\partial x^j} - e_i^j e_m^\ell \Gamma_{j\ell}^k(x) \frac{\partial}{\partial e_m^k}.$$

So for $U_t = (X_t^i, e_j^i(t))$,

$$\begin{aligned} dX_t^i &= e_j^i(t) \circ dW_t^j, \\ de_j^i(t) &= -\Gamma_{k\ell}^i(X_t) e_j^\ell(t) e_m^k(t) \circ dW_t^m. \end{aligned}$$

For the first equation,

$$dX_t^i = e_j^i(t) dW_t^j + \frac{1}{2} d\langle e_j^i, dW^j \rangle_t.$$

Let

$$dM_t^i = e_j^i(t) dW_t^j \Rightarrow d\langle M^i, M^j \rangle_t = \sum_{k=1}^d e_k^i(t) e_k^j(t) dt.$$

Note that because $u \in O(M)$ and $ue_\ell = e_\ell^i \frac{\partial}{\partial x^i}$,

$$\delta_{\ell m} = \langle ue_\ell, ue_m \rangle = e_\ell^i g_{ij} e_m^j,$$

i.e., $ege^\top = I$ for $e = (e_j^i)$. Moreover, because e is invertible,

$$e^\top = g^{-1}, \quad \text{i.e.} \quad \sum_{k=1}^d e_k^i e_k^j = g^{ij}.$$

Therefore,

$$d\langle M^i, M^j \rangle_t = g^{ij}(X_t) dt.$$

Then let $\sigma = g^{\frac{1}{2}}$ and define

$$B_t = \int_0^t \sigma(X_s)^{-1} dM_s,$$

Because

$$d\langle B^i, B^j \rangle_t = \delta^{ij} dt,$$

B is an Euclidean Brownian motion. Also,

$$dM_t = \sigma(X_t) dB_t.$$

On the other hand, by the second equation,

$$d\langle e_j^i, dW^j \rangle_t = -\Gamma_{k\ell}^i(X_t) e_m^\ell(t) e_m^k(t) = -g^{\ell k}(X_t) \Gamma_{k\ell}^i(X_t).$$

Therefore, X_t satisfies

$$dX_t^i = \sigma_j^i(X_t) dB_t^j - \frac{1}{2} g^{\ell k}(X_t) \Gamma_{k\ell}^i(X_t) dt,$$

where B is a d -dimensional Brownian motion. Note that X_t is a diffusion process generated by

$$L = -\frac{1}{2} g^{\ell k} \Gamma_{k\ell}^i \frac{\partial f}{\partial x^i} + \frac{1}{2} g^{ij} \frac{\partial^2 f}{\partial x^i \partial x^j}.$$

Also, because

$$\Delta_M f = g^{ij} \frac{\partial^2 f}{\partial x^i \partial x^j} - g^{k\ell} \Gamma_{k\ell}^i \frac{\partial f}{\partial x^i} = 2L,$$

X is a $\Delta_M/2$ -diffusion process, and so X is a Brownian motion on M .

3.4 Radial Process

Let (M, g) be a complete Riemannian manifold of dimension d . Fix $o \in M$ and let C_o be the cut locus and $E_o = M \setminus C_o$. Let

$$r(x) = d(x, o).$$

Then r is smooth on $M \setminus C_o \cup \{o\}$ and $|\nabla r| \equiv 1$.

Let X be a Brownian motion on M starting within E_o . Before it hitting C_o , i.e., $t < T_{C_o}$, where T_{C_o} is the hitting time of C_o ,

$$\beta_t = r(X_t) - r(X_0) - \frac{1}{2} \int_0^t \Delta_M r(X_s) ds.$$

is a local martingale. Moreover, because

$$\langle r, r \rangle_t = \langle \beta, \beta \rangle_t = \int_0^t \Gamma(r, r)(X_s) ds = \int_0^t |\nabla r(X_s)|^2 ds = t,$$

$r(X)$ is a Brownian motion on $\mathbb{R}$.

Model Manifolds. Choose the polar coordinate (r, θ) of $\mathbb{R}^n$, then equip $\mathbb{R}^n$ with Riemannian metric

$$g = dr \otimes dr + G(r)^2 g_{\mathbb{S}^{n-1}}.$$

where $G: [0, \infty) \rightarrow [0, \infty)$ is smooth with $G(0) = 0$, $G'(0) = 1$, and $G(r) > 0$ for $r \neq 0$. Then any Riemannian manifold $(\mathbb{M}, g)$ isometric to $(\mathbb{R}^n, g)$ is called a model manifold, and denote $0 \in \mathbb{M}$ be a fixed point corresponding to $0 \in \mathbb{R}^n$. For the Levi-Civita connection ∇ , we have the Koszul formula

$$\begin{aligned} 2 \langle \nabla_U V, W \rangle &= U \langle V, W \rangle + V \langle W, U \rangle - W \langle U, V \rangle \\ &\quad + \langle [U, V], W \rangle - \langle [V, W], U \rangle - \langle [U, W], V \rangle. \end{aligned}$$

So, let $X_r = \frac{\partial}{\partial r}$ and $Y, Z \in \Gamma(T\mathbb{S}^{n-1})$, because

$$\langle X_r, X_r \rangle = 1, \quad \langle Y, Z \rangle = G(r)^2 \langle Y, Z \rangle_{\mathbb{S}^{n-1}}, \quad [X_r, Y] = [X_r, Z] = 0,$$

we have the following calculations.

(1) First, because $\langle X_r, X_r \rangle = 1$,

$$0 = X_r \langle X_r, X_r \rangle = 2 \langle \nabla_{X_r} X_r, X_r \rangle = 0.$$

So $\nabla_{X_r} X_r \in \Gamma(T\mathbb{S}^{n-1})$. On the other hand, because $Y \perp X_r$, by Koszul formula,

$$2 \langle \nabla_{X_r} X_r, Y \rangle = -Y \langle X_r, X_r \rangle = 0.$$

Therefore,

$$\nabla_{X_r} X_r = 0.$$

(2) By Koszul formula, $\langle \nabla_{X_r} Y, X_r \rangle = 0$, which implies that $\nabla_{X_r} Y \in \Gamma(T\mathbb{S}^{n-1})$. Similarly,

$$2 \langle \nabla_{X_r} Y, Z \rangle = X_r (\langle Y, Z \rangle) = 2G(r)G'(r) \langle Y, Z \rangle_{\mathbb{S}^{n-1}} = 2 \frac{G'(r)}{G(r)} (\langle Y, Z \rangle).$$

Therefore,

$$\nabla_{X_r} Y = \frac{G'(r)}{G(r)} Y.$$

By torsion-free,

$$\nabla_Y X_r = [X_r, Y] + \nabla_{X_r} Y = \nabla_{X_r} Y = \frac{G'(r)}{G(r)} Y.$$

(3) Also, by Koszul formula,

$$\langle \nabla_Y Z, X_r \rangle = -\frac{G'(r)}{G(r)} \langle Y, Z \rangle.$$

Therefore,

$$\nabla_Y Z = \nabla_Y^{\mathbb{S}^{n-1}} Z - \frac{G'(r)}{G(r)} \langle Y, Z \rangle X_r,$$

where $\nabla^{\mathbb{S}^{n-1}}$ is the induced Levi-Civita connection.

The radical curvature κ_{rad} is defined as the sectional curvature restricted to radical section, that is a $\Pi_x \subset T_x M$, where $\Pi_x = \text{span}\{X_r, Y\}$ for some unit $Y \in \Gamma(T\mathbb{S}^{n-1})$. To calculate the radical curvature, by above, we have

$$\nabla_Y Y = \nabla_Y^{\mathbb{S}^{n-1}} Y - \frac{G'(r)}{G(r)} X_r,$$

which implies that

$$\nabla_{X_r} \nabla_Y Y = \frac{G'}{G} \nabla_Y^{\mathbb{S}^{n-1}} Y - \frac{G''G - (G')^2}{G^2} X_r,$$

and

$$\begin{aligned} \nabla_Y \nabla_{X_r} Y &= \nabla_Y \left(\frac{G'}{G} Y \right) = \frac{G'}{G} \nabla_Y Y \\ &= \frac{G'}{G} \nabla_Y^{\mathbb{S}^{n-1}} Y - \left(\frac{G'}{G} \right)^2 X_r. \end{aligned}$$

Therefore,

$$\begin{aligned} R(X_r, Y)Y &= \nabla_{X_r} \nabla_Y Y - \nabla_Y \nabla_{X_r} Y - \nabla_{[X_r, Y]} Y \\ &= \nabla_{X_r} \nabla_Y Y - \nabla_Y \nabla_{X_r} Y \\ &= -\frac{G''}{G} X_r. \end{aligned}$$

It follows that

$$K(\Pi_x) = \frac{\langle R(X_r, Y)Y, X_r \rangle}{|X_r|^2 |Y|^2 - \langle X_r, Y \rangle^2} = -\frac{G''(r)}{G(r)},$$

where $r = d(0, x)$. Note that the radial curvature is independent of the choice of Y , i.e.

$$\kappa_{\text{rad}}(r) = -\frac{G''(r)}{G(r)}.$$

i.e., G is the unique solution of ODE

$$G''(r) + \kappa_{\text{rad}}(r)G(r) = 0, \quad G(0) = 0, \quad G'(0) = 1.$$

Moreover,

$$\text{Ric}(X_r, X_r) = -(n-1) \frac{G''(r)}{G(r)}$$

Then let $\gamma(t) = (t, \theta_0)$ be a normal radial geodesic with $\dot{\gamma}(t) = X_r(t)$. Let $E_1 = X_r, E_2, \dots, E_n$ be a orthonormal frame along γ . If

$$J(t) = f(t)E(t)$$

is an orthogonal Jacobian field, then f satisfies

$$f''(t) + \kappa_{\text{rad}}(t)f(t) = 0.$$

Therefore, for any $X \in T_{\gamma(t_0)}\mathbb{S}^{n-1}$, let $J(t)$ be the orthogonal Jacobian field with $J(0) = 0$ and $J(t_0) = X$. Note that for such $J(t) = f(t)E(t)$,

$$f''(t) + \kappa_{\text{rad}}(t)f(t) = 0, \quad f(0) = 0,$$

which implies that $f(t) = cG(t)$ for some constant c . Then

$$\nabla^2 r(X, X) = \langle \nabla_{X_r} J(t_0), J(t_0) \rangle = \frac{f'(t_0)}{f(t_0)} |X|^2 = \frac{G'(t_0)}{G(t_0)} |X|^2.$$

Therefore,

$$\Delta r(t) = \sum_{i=2}^{\infty} \nabla^2 r(E_i, E_i) = (n-1) \frac{G'(t_0)}{G(t_0)}.$$

Generalize Comparison Theorem. Let

$$\begin{aligned} K_M(x) &= \{K(\Pi): \Pi \subset T_x M \text{ 2-dimensional section}\}, \\ \text{Ric}_M(x) &= \{\text{Ric}(X, X): X \in T_x M, |X| = 1\}. \end{aligned}$$

Assume there are two model manifolds with radial curvature κ_1 and κ_2 such that

$$\begin{aligned} \kappa_1(r) &\geq \sup \{K_M(x): r(x) = r\} \\ \kappa_2(r) &\leq \inf \{\text{Ric}_M(x): r(x) = r\} (d-1)^{-1}. \end{aligned}$$

Then by replacing the space form with the model manifold $(\mathbb{M}, g)$ in the proof of Hessian comparison theorem and Laplacian comparison theorem, we have the following comparison,

$$(d-1) \frac{G'_1(r)}{G_1(r)} \leq \Delta_M r \leq (d-1) \frac{G'_2(r)}{G_2(r)},$$

where

$$G''(r) + \kappa(r)G(r) = 0, \quad G(0) = 0, \quad G'(0) = 1.$$

Chapter 4

Riemannian Langevin Dynamics

Let (M, g) be a complete Riemannian manifold with Levi-Civita connection ∇ and the canonical Riemannian volume measure dm . In the following, we denote $\text{grad } f = \nabla f$ for convenience. Let B^M is a Brownian motion on M . Let $U \in C^\infty(M)$. Consider the following SDE

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}dB_t^M, \quad (4.1)$$

which is called a Riemannian Langevin dynamics.

4.1 Diffusion Operator

Because B^M is a Brownian motion on M , its anti-development W is a standard Euclidean Brownian motion. So

$$dB_t^M = V_\alpha \circ dW_t^\alpha,$$

with

$$\Delta_M = \sum_{\alpha} V_\alpha^2.$$

Therefore,

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}V_\alpha \circ dW_t^\alpha,$$

which implies that the diffusion operator of X is

$$\begin{aligned} Lf &= \frac{1}{2} \sum_{\alpha} (\sqrt{2}V_\alpha)^2 f - \nabla U(f) \\ &= \Delta_M f - \langle \nabla U, \nabla f \rangle, \end{aligned}$$

i.e., $L = \Delta_M - \langle \nabla U, \nabla \cdot \rangle$. For such L , in note *Stochastic Analysis*, we have shown that

$$L^*p = \Delta_M p + \text{div}(p \nabla U)$$

in the sense of on $L^2(dm)$. Also,

$$d\mu = e^{-U} dm,$$

is the invariant measure of such diffusion process, which is also symmetric for L . Note that L is essentially self-adjoint on $C_c(M)$ w.s.t. μ . Therefore, it has a unique self-adjoint extension L on $L^2(\mu, M)$. Let

$$P_t = e^{-tL}$$

be the functional calculus of L , defined on $L^2(\mu, M)$. On the other hand, define

$$Q_t f(x) = \mathbb{E}[f(X_t) \mid X_0 = x], \quad \forall f \in C_c^\infty(M),$$

By note *Markov Operator*, Q_t can be extended to $L^2(\mu)$ and has the generator A that is self-adjoint on $L^2(\mu)$. By Dynkin's formula,

$$Af = Lf, \quad \forall f \in C_c^\infty(M).$$

Therefore, $L \subset A$. By the essential self-adjointness of L , $L = A$, which implies that

$$P_t f(x) = Q_t f(x) = \mathbb{E}[f(X_t) \mid X_0 = x], \quad \forall f \in L^2(\mu).$$

Remark 4.1.1. Note that if ∇U is globally Lipschitz continuous (in particular, M is compact), i.e., $|\nabla^2 U| \leq L$, then it can show that $(Q_t)_{t \geq 0}$ is a Feller-Dynkin semigroup so that it can be defined on $C_0(M)$.

Remark 4.1.2. For P_t , note that

$$\frac{\partial}{\partial t} P_t = L P_t = P_t L,$$

called the backward Kolmogorov equation. Moreover, it can define the Markov transition kernel

$$p_t(x, A) = P_t \mathbb{I}_A(x) = \mathbb{E}[\mathbb{I}_A(X_t) \mid X_0 = x] = \mathbb{P}_x(X_t \in A).$$

Then

$$P_t f(x) = \int_M f(y) p_t(x, dy).$$

It follows that we can define

$$\int_M f d(P_t^* \nu) = \int_M P_t f d\nu.$$

Then we have

$$\frac{\partial}{\partial t} P_t^* = L^* P_t^* = P_t^* L^*,$$

called the forward Kolmogorov equation/Fokker-Planck equation, where L^* acts on ν by

$$\int_M f d(L^* \nu) = \int_M L f d\nu.$$

4.2 Deterministic Flow

Let $V(t, x)$ be a time-dependent C^∞ -vector field. Consider ODE on M

$$\begin{cases} \frac{dx_t}{dt} = V(t, x_t) \\ x_0 = x \end{cases} \quad (4.2)$$

and denote $\varphi_t(x)$ be its flow.

Theorem 4.2.1. *Let $\{\varphi_t\}_{t \in [0, T]}$ be a locally Lipschitz family of diffeomorphisms of M with $\varphi_0 = id$. Let $V(t, x)$ be the time-dependent vector fields associated with $\{\varphi_t\}_{t \in [0, T]}$. Let ν be a probability measure on M and $\nu_t = (\varphi_t)_\# \nu$. Then ν_t is the unique solution of the following PDE*

$$\begin{cases} \frac{\partial \nu_t}{\partial t} + \operatorname{div}(\nu_t V_t) = 0, & t \in [0, T) \\ \nu_0 = \nu \end{cases} \quad (4.3)$$

in $C([0, T], \mathcal{P}(M))$, where $\mathcal{P}(M)$ is equipped with the weak topology.

Remark 4.2.2. For the measure PDE, it means that for any $f \in C_c^\infty(M)$,

$$\langle \partial_t \nu_t, f \rangle + \langle \operatorname{div}(\nu_t V_t), f \rangle = 0,$$

where

$$\begin{aligned} \langle \partial_t \nu_t, f \rangle &= \frac{\partial}{\partial t} \langle \nu_t, f \rangle = \frac{\partial}{\partial t} \int_M f d\nu_t, \\ \langle \operatorname{div}(\nu_t V_t), f \rangle &= - \int_M f d(\operatorname{div}(\nu_t V_t)) = - \int_M \operatorname{div}(f V_t) d\nu_t = \int_M \langle \nabla f, V_t \rangle d\nu_t. \end{aligned}$$

Moreover, if $d\nu_0 = p dm$ for $p \in C^\infty$, which implies there are $p_t \in C^\infty$ such that $d\nu_t = p_t dm$, then the PDE is equivalent to

$$\begin{cases} \frac{\partial p_t}{\partial t} + \operatorname{div}(p_t V_t) = 0, & t \in [0, T) \\ p_0 = p. \end{cases}$$

The theorem implies that if there is a family of probability measures $\{\nu_t\}_{t \in [0, T]}$ satisfies above PDE (4.3), then for a random process $\tilde{X}_t$ on M satisfying the ODE (4.2) with $\tilde{X}_0 \sim \nu$, we have $\tilde{X}_t \sim \nu_t$.

Consider the Riemannian Langevin dynamics (4.1) with $X_0 \sim \nu$, where $d\nu = p dm$ for a smooth p . Then $X_t \sim \nu_t$, for $d\nu_t = d(P_t^* \nu) = p_t dm$. By the Fokker-Planck equation, we have

$$\frac{\partial \nu_t}{\partial t} = L^* \nu_t,$$

which is equivalent to

$$\frac{\partial p_t}{\partial t} = \Delta_M p_t + \operatorname{div}(p_t \nabla U) = \operatorname{div}(p_t (\nabla U + \nabla \log p_t)).$$

Therefore, if $\tilde{X}_t$ satisfies

$$\begin{cases} \frac{d\tilde{X}_t}{dt} = -\nabla U(\tilde{X}_t) - \nabla \log p_t(\tilde{X}_t), \\ \tilde{X}_0 \sim \nu, \end{cases}$$

then $\tilde{X}_t \sim \nu_t$, i.e., $\tilde{X}_t \stackrel{d}{=} X_t$.

4.3 Convergence to Invariant Measure

KL Divergence. If we have

$$\operatorname{Ric}_U = \operatorname{Ric} + \nabla^2 U \geq \rho g,$$

for some $\rho > 0$, then above process satisfies the $\operatorname{CD}(\rho, \infty)$ and so $\operatorname{LSI}(1/\rho)$, which implies that

$$\operatorname{Ent}_\mu(P_t g) \leq e^{-2\rho t} \operatorname{Ent}_\mu(g), \quad \forall g \in L^1(\mu).$$

(see note *Stochastic Analysis*). If $X_0 \sim \nu_0$ with $d\nu_0 = f d\mu$ (assume $f \in L^1(\mu)$) and $\operatorname{KL}(\nu_0 \parallel \mu) < \infty$, then

$$X_t \sim \nu_t = P_t^* \nu_0,$$

where $d(P_t^* \nu_0) = P_t f d\mu$. Then we can get

$$\operatorname{KL}(\nu_t \parallel \mu) \leq e^{-2\rho t} \operatorname{KL}(\nu_0 \parallel \mu).$$

Therefore,

$$\operatorname{KL}(\nu_t \parallel \mu) \leq \mathcal{O}(e^{-t}).$$

On the other hand, by Pinsker inequality $d_{\operatorname{TV}}(\nu_t, \mu) \leq \frac{1}{2} \operatorname{KL}(\nu_t \parallel \mu)$,

$$d_{\operatorname{TV}}(\nu_t, \mu) \leq \mathcal{O}(e^{-t}).$$

Wasserstein Distance. Under the same condition,

$$\mathrm{Ric}_U = \mathrm{Ric} + \nabla^2 U \geq \rho g,$$

we can have

$$W_2(\nu_t, \mu) \leq e^{-\rho t} W_2(\nu_0, \mu).$$

Remark 4.3.1. It needs techniques from the optimal transport theory, like W_2 -geodesic convexity, et. al., or it can be obtained by directly calculating from the SDE.